

Computational Limits and Robustness of Predicting Developmental Cell-Fate Landscapes: Three Studies on Real Data

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Abstract

When can computation predict developmental cell-fate landscapes, and at what cost in information and robustness? We answer across three studies on real, public, accession-level data, organized as one argument: (A) emergent pattern formation resists image-based prediction, and a physics check on measured transport parameters shows *where* self-organization can operate at all; (B) fate robustness depends on transition dynamics, perturbation kernel and starting measure, with ranking reversals in a published floral model; (C) positional states are decodable far below published information budgets. Study A simulates Turing reaction–diffusion morphogen patterning (Gierer–Meinhardt kinetics) and measures a systematic, statistically robust *beyond-onset wavenumber upshift*: simulated patterns select wavenumbers 15–60% above the linear fastest-growing mode, with the largest deviations at intermediate distance from onset; we show honestly that this shift is *not* predictable from the linear dispersion relation by a convolutional surrogate ($n = 171$ simulations). Study B analyses the published *Drosophila* segment-polarity Boolean network (Cell Collective id-021) by exact attractor enumeration under all eight external signalling conditions, reproduces the published bistability, quantifies the wild-type basin as 0.77% of the synchronous state space, and measures the sample complexity of learning the basin boundary (graph neural network vs. multilayer perceptron). Study C analyses 676 real Bicoid-GFP morphogen gradients and 2,180 gap-gene immunofluorescence embryo records (Liu, Morrison & Gregor, PNAS 2013): we (i) reproduce the gradient length scale under the authors’ own fitting window, (ii) quantify a dosage-coupled length-scale effect that is concentrated in the anterior embryo (Spearman $\rho = 0.36$, $p = 8 \times 10^{-16}$ in anterior windows; absent in the authors’ posterior window), (iii) beat the published threshold model of cephalic-furrow positioning with a one-dimensional convolutional decoder (RMSE 2.38% egg length vs. 4.94% egg length, held-out fly lines; paired bootstrap CI for the gain [1.99, 3.11]% EL),

and (iv) show that positional decodability from Hb+Eve expression improves with maternal *bcd* dosage (6.21% vs. 9.91% EL median error; bootstrap CI [3.00, 4.40]% EL). Two headline measurements tie the studies together: a *developmental fragility index* for the segment-polarity network ($F = 0.500$; exactly half of all 1,792 single-node perturbations of wild-type-basin states destroy the wild-type outcome, concentrated entirely in the CI/EN axis, with seven clamp interventions that abolish the basin and six that double it) and a *minimum-information threshold* for morphogen readout (cephalic-furrow position is decodable below the published benchmark from a single measured point of the Bicoid gradient, 4.29% vs 4.94% EL). Negative results (a retracted small-sample surrogate win, single-exponential sufficiency, degenerate basin classification) are reported alongside positive ones, each with its positive interpretation. All code, tests, data manifests and result files are public at the project repository.

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1 Introduction

1.1 Digital embryos

A *digital embryo* is a computational reconstruction of a developmental process that is close enough to the underlying biology to be wrong in interesting ways. The promise of the field is that cheap computation can replace some expensive experimentation: simulate the morphogen, the network, or the tissue, and reserve the bench for questions the simulation cannot answer. The peril is equally well known — a simulation unconstrained by real measurements will happily reproduce whatever its author expected. This report takes the second half of that sentence seriously. Three classical model systems of developmental patterning are rebuilt computationally, and every quantitative claim is tied to real, accession-level public data: published per-embryo morphogen measurements, a published gene-regulatory network model, and published quantitative benchmarks. Where our computation contradicts an initial expectation — including our own earlier, stronger claims — we say so in the body of the report, not in a footnote.

The three studies correspond to the three great patterning problems of the fly embryo, at three scales:

1. **Turing pattern selection (Study A).** Reaction–diffusion systems of the type proposed by Turing [1] and instantiated by Gierer and Meinhardt [2] amplify a band of spatial wavelengths out of noise. Linear theory predicts which wavenumber grows fastest; what the *saturated, nonlinear* pattern actually selects is a subtler question. We measure the selected wavenumber across a parameter sweep and ask whether the deviation from linear theory is itself predictable.
2. **Gene-network attractors (Study B).** The segment-polarity network of *Drosophila* is one of the best-studied Boolean models in systems biology [8, 9]. We take the published 14-node Boolean model (Cell Collective id-021 [10]) as ground truth, enumerate its attractors exactly under all eight external signalling conditions, measure the size of the wild-type basin of attraction, and ask how much data a learning algorithm needs to recover the basin boundary.
3. **Morphogen positional information (Study C).** The Bicoid morphogen gradient of the early fly embryo is the canonical example of positional information [3, 4]. Using the 676 per-embryo gradients and 2,180 immunofluorescence embryo records published by Liu, Morrison and Gregor [5], we (i) reproduce their gradient length-scale measurements, (ii) test how the length scale depends on maternal gene dosage, (iii) attack their threshold model of cephalic-furrow positioning with a learned decoder, and (iv) measure how much positional information gap-gene expression patterns carry under different dosages.

1.2 Why these three

The three problems are independent in method but connected in spirit: each asks where spatial order comes from, and each has a quantitative published reference point that a computation can fail to meet. Study A’s reference is the linear dispersion relation itself — a prediction made by mathematics, not by a measurement, which makes it the cleanest possible benchmark to break. Study B’s reference is a published, curated, executable model: if our attractor enumeration disagreed with it, the disagreement would be diagnosable down to a single update rule. Study C’s references are the hardest kind: numbers measured on real embryos, with real noise, by another laboratory [5, 6, 4].

1.3 Summary of results

Every number quoted here is computed by committed, tested code from committed data, and re-computed when this report is built; the repository records the exact pipeline. Section references point to full derivations and tables.

- **Study A (Sections 3).** Across a 14-point sweep of the activator decay rate, the saturated Gierer–Meinhardt pattern systematically selects a wavenumber *above* the linear fastest-growing mode, by 14.7% to 58.9% (median 31%), with the largest relative deviations at intermediate distance from onset — not at threshold, where amplitude-equation reasoning would have placed them. The deviation is real (measured at steady state, verified against two lattice sizes) but it is *not* predictable from the linear dispersion curve by a convolutional surrogate: a promising $n = 58$ pilot result (+17.7% held-out improvement) *failed to replicate* at $n = 171$ (CNN relative RMSE 0.292–0.295 vs. 0.272 for the linear formula alone). We retract the small-sample claim and keep the phenomenon.
- **Study B (Section 4).** Exact attractor enumeration of the published segment-polarity network under all eight combinations of the external signals HH, WG, SLP reproduces the published condition-dependent bistability exactly [9]. The wild-type (wg-ON) basin occupies 0.77% of the synchronous state space under the wild-type signalling condition. The basin boundary is learnable to $> 98\%$ test accuracy from only ~ 60 labelled states (of 2^{14}), and a graph neural network matches or beats a multilayer perceptron at every sample size; under naive uniform sampling the classification task is degenerate (majority class accuracy 99.3%), which we report as a cautionary result.
- **Study C (Sections 5–5.7).** From 676 real Bcd-GFP gradients: the population length scale reproduces the published value under the authors’ fitting window ($\lambda_{\text{median}} = 0.189$ vs. published 0.165 ± 0.007 egg lengths [5], with our value on their exact window; the residual dif-

ference is discussed honestly). The length scale *increases with maternal bcd dosage* (dosage-coupled length scale, DCLS): Spearman $\rho = 0.356$ ($p = 7.9 \times 10^{-16}$) on anterior-weighted fitting windows, robust to parametric refitting (OLS slope +0.0117 per dose, 95% CI [0.0060, 0.0174], $p = 5.9 \times 10^{-5}$), and *absent* on the posterior-heavy window used in the original publication ($\rho = 0.071$, $p = 0.12$) — a window-sensitivity finding, not a contradiction. A one-dimensional convolutional decoder predicts cephalic-furrow position from the raw Bcd profile at 2.38% egg length RMSE (held-out fly lines, $R^2 = 0.854$), beating the published threshold model at 4.94% EL by a paired-bootstrap margin of 2.57% EL, CI₉₅ [1.99, 3.11] ($p < 10^{-4}$). Finally, positional decodability from joint Hb+Eve expression improves with dosage: median error 6.21% EL at $3\times$ bcd vs. 9.91% EL at $1\times$, bootstrap CI [3.00, 4.40]% EL for the difference.

1.4 What counts as evidence here

This project operates under a simple reporting rule: a number is quoted only if it exists in a committed results file produced by committed, tested code; results that fail to replicate are retracted in the same document that reports them; and the negative results (Section 8) are given the same space and the same figures as the positive ones. We regard the retracted CNN-surrogate claim of Study A as the single most valuable output of the project, because it is the kind of result that a less careful pipeline would have published.

1.5 Background

Turing patterns. Turing’s 1952 observation [1] — that diffusion, usually a homogenising force, can destabilise a homogeneous state when one morphogen activates and another, faster-diffusing one inhibits — is the founding result of mathematical developmental biology. The linear theory is textbook material; the nonlinear selection problem it leaves open (which mode, of the whole unstable band, does the saturated pattern actually take?) has a large literature for specific kinetics and geometries but no general closed answer. Study A measures the gap between linear prediction and nonlinear outcome on the canonical activator–inhibitor kinetics of Gierer and Meinhardt [2].

The segment-polarity network. The repeated unit of the embryonic parasegment is maintained by a small network of cross-cell signalling (wingless, hedgehog) and intracellular regulation (engrailed, cubitus interruptus, patched). von Dassow et al. showed by exhaustive ODE sampling that the module is extraordinarily robust to kinetic parameters [8]; Albert and Othmer then showed that a Boolean skeleton of the same network already contains the essential logic [9]. The curated Boolean version we use [10] is therefore a model with an unusually strong claim to correctness — which is exactly why exact numbers about it (basin sizes, learnability) are worth having.

Positional information in the fly embryo. Wolpert’s positional information framework [3] proposes that cells read their position from morphogen concentration. The Bicoid gradient is its best quantitative instance: an approximately exponential anterior morphogen gradient [4] that is read out by gap genes with a precision approaching the physical limit set by expression noise [7, 6]. Liu, Morrison and Gregor added the crucial dosage series — embryos with $1\times$ to $4\times$ maternal *bcd* — and showed the readout machinery tracks the input dynamically [5]. Study C is built entirely on their public data.

1.6 Roadmap

Section 2.1ff. derives every formula used later, with symbolic cross-checks. Sections 3, 4 and 5 present the three studies, each self-contained in method and result. Section 6.6 documents the software, including the *embryosim* command-line tool. Section 7 discusses scope and open work; Section 8 collects the negative results. Appendices carry the full tool and dataset manifests (Appendix A), the complete attractor census (Appendix B), reproduction commands (Appendix C), the results-file inventory (Appendix D) and the test-suite inventory (Appendix E).

2 Mathematical preliminaries and derivations

All computations in this report rest on a small set of mathematical objects, derived here in full so that every subsequent claim can be traced to an explicit formula. Each derivation below is additionally checked *symbolically* by computer algebra (sympy 1.14) in the repository test suite (`tests/test_math_verify.py`), which asserts exact equality between the symbolic results quoted here and the numerical values committed in `results/results.json`. Where a formula is later used to produce a number, the section producing that number says so explicitly.

2.1 Turing systems: linear stability of the homogeneous state

We consider two-morphogen reaction–diffusion systems on a square domain $\Omega = [0, L]^2$ with periodic boundary conditions,

$$\partial_t a = f(a, h) + D_a \nabla^2 a, \quad \partial_t h = g(a, h) + D_h \nabla^2 h, \quad (1)$$

where $a(\mathbf{x}, t)$ and $h(\mathbf{x}, t)$ are the activator and inhibitor concentrations, $D_a, D_h > 0$ their diffusion coefficients, and f, g the local reaction kinetics. The specific kinetics used throughout Study A are the rescaled Gierer–Meinhardt kinetics [2],

$$f(a, h) = \rho \frac{a^2}{h} - \mu_a a, \quad g(a, h) = \rho a^2 - \mu_h h, \quad (2)$$

with source density $\rho > 0$ and decay rates $\mu_a, \mu_h > 0$.

Proposition 1 (Homogeneous steady state). *The system (1)–(2) has a unique spatially homogeneous steady state with $a^* > 0$, namely*

$$a^* = \frac{\mu_h}{\mu_a}, \quad h^* = \frac{\rho \mu_h}{\mu_a^2}. \quad (3)$$

Proof. Setting $f = 0$ with $a > 0$ gives $\rho a^2/h = \mu_a a$, i.e. $h = \rho a/\mu_a$. Setting $g = 0$ gives $h = \rho a^2/\mu_h$. Eliminating h yields $\rho a/\mu_a = \rho a^2/\mu_h$, hence $a^* = \mu_h/\mu_a$; back-substitution gives h^* . Symbolic check: `sympy.solve` returns exactly this pair and no other positive solution (repository `test test_gm_steady_state`). $\square$

Linearising about (a^*, h^*) with $u = a - a^*, v = h - h^*$ gives $\partial_t(u, v)^\top = J(u, v)^\top + D\nabla^2(u, v)^\top$ with $D = \text{diag}(D_a, D_h)$ and Jacobian

$$J = \begin{pmatrix} f_a & f_h \\ g_a & g_h \end{pmatrix}_{(a^*, h^*)} = \begin{pmatrix} \mu_a & -\mu_a^2/\mu_h \\ 2\mu_h & -\mu_h \end{pmatrix}, \quad \text{tr } J = \mu_a - \mu_h, \quad \det J = \mu_a \mu_h. \quad (4)$$

(The four entries follow from $f_a = 2\rho a/h - \mu_a$, $f_h = -\rho a^2/h^2$, $g_a = 2\rho a$, $g_h = -\mu_h$ evaluated at (3); symbolic verification in the test suite.) For the standard parameter set used in our validation run ($\mu_a = 0.06$, $\mu_h = 0.12$, $\rho = 1$) this gives $a^* = 2$, $h^* = 33.\bar{3}$, $\text{tr } J = -0.06$, $\det J = 0.0072$ — identical to the values committed in `results/results.json`.

Because the Laplacian diagonalises over Fourier modes $e^{i\mathbf{k}\cdot\mathbf{x}}$, each wavenumber $\mathbf{k}$ evolves independently with matrix $M(k^2) = J - k^2 D$, $k^2 = |\mathbf{k}|^2$. Its eigenvalues $\sigma_\pm(k^2)$ satisfy

$$\sigma^2 - \tau(k^2)\sigma + \Delta(k^2) = 0, \quad \tau(k^2) = \text{tr } J - (D_a + D_h)k^2, \quad \Delta(k^2) = \det J - Bk^2 + D_a D_h k^4, \quad (5)$$

where

$$B = D_h f_a + D_a g_h = D_h \mu_a - D_a \mu_h. \quad (6)$$

Since $\tau(k^2) < 0$ whenever $\text{tr } J < 0$, an instability of the homogeneous state to pattern formation requires $\Delta(k^2) < 0$ for some $k^2 > 0$ (the Turing condition [1]). As Δ is an upward parabola in k^2 , the unstable band is the interval between its two positive roots,

$$k_\pm^2 = \frac{B \pm \sqrt{B^2 - 4D_a D_h \det J}}{2D_a D_h}, \quad \text{real and positive iff } B > 2\sqrt{D_a D_h \det J}, \quad (7)$$

the latter being the classic condition $D_h \mu_a - D_a \mu_h > 2\sqrt{D_a D_h \mu_a \mu_h}$, i.e. the inhibitor must diffuse sufficiently faster than the activator. The dispersion peak (fastest-growing mode) sits at the vertex of the parabola,

$$k_{\max}^2 = \arg \max_{k^2} \text{Re } \sigma_+(k^2) = \frac{B}{2D_a D_h} = \frac{\mu_a}{2D_a} - \frac{\mu_h}{2D_h}, \quad (8)$$

where the second equality is exact whenever τ varies negligibly across the band; the symbolic derivation in the test suite confirms that $\partial \operatorname{Re} \sigma_+ / \partial k^2 = 0$ at this point to leading order, and our simulations use (8) as the linear prediction k_{lin}^2 . For the validation parameter set, (7) gives $k_-^2 = 0.671$, $k_+^2 = 10.729$, exactly the committed values, and $k_{\text{max}}^2 = 5.70$.

Definition 1 (Distance from onset). *With μ_a swept at fixed μ_h, D_a, D_h , the onset value μ_a^{crit} is the largest μ_a for which the instability condition in (7) holds, i.e. the root of $B = 2\sqrt{D_a D_h \det J}$. We measure distance from onset by*

$$\varepsilon = 1 - \frac{\mu_a}{\mu_a^{\text{crit}}}. \quad (9)$$

For our sweep ($D_a = 0.005$, $D_h = 0.2$, $\mu_h = 0.12$) one finds $\mu_a^{\text{crit}} = 0.12$, so $\varepsilon = 1 - \mu_a/0.12$.

2.2 Finite-difference integration and the stability bound

We discretise (1) on an $n \times n$ lattice with spacing $\Delta x = 1$ and the five-point Laplacian, and integrate with explicit Euler steps. The Laplacian eigenvalues on the lattice lie in $[-8, 0]$ (for unit spacing, the five-point stencil with periodic boundaries), and the explicit diffusion update $u \mapsto u + \Delta t D \Delta_h u$ is stable only if $|1 - 8\Delta t D| \leq 1$ in the worst mode, giving the CFL-type bound

$$\Delta t \leq \frac{1}{4D_{\text{max}}}, \quad D_{\text{max}} = \max(D_a, D_h). \quad (10)$$

All simulations use $\Delta t = 0.8/(4D_{\text{max}})$, enforced in code with a hard error when violated (GiererMeinhardt2D...). The discrete dispersion relation replaces k^2 by the lattice symbol $\tilde{k}^2 = 4\sin^2(k_x/2) + 4\sin^2(k_y/2)$; the difference between $\tilde{k}^2$ and k^2 is one of the honest sources of wavenumber quantisation we track in Study A (the “lattice-rounded linear mode” of Section 3).

2.3 Morphogen gradients: synthesis–diffusion–degradation

The standard model of a maternally deposited morphogen such as Bicoid (Bcd) is the synthesis–diffusion–degradation (SDD) equation [4]: protein is produced at the anterior pole at flux J_0 , diffuses with coefficient D , and is degraded uniformly at rate γ . In the steady state, on a one-dimensional axis $x \geq 0$ (fractional egg length, $x = 0$ anterior),

$$D c''(x) = \gamma c(x), \quad -D c'(0) = J_0, \quad c(\infty) = 0. \quad (11)$$

Proposition 2 (Exponential gradient). *The unique bounded solution of (11) is*

$$c(x) = \frac{J_0}{\sqrt{D\gamma}} e^{-x/\lambda}, \quad \lambda = \sqrt{\frac{D}{\gamma}}, \quad (12)$$

i.e. an exponential with length scale λ .

Proof. The general solution of $Dc'' = \gamma c$ is $c = C_1 e^{x/\lambda} + C_2 e^{-x/\lambda}$ with $\lambda = \sqrt{D/\gamma}$; boundedness forces $C_1 = 0$, and the flux condition $-Dc'(0) = DC_2/\lambda = J_0$ fixes $C_2 = J_0\lambda/D = J_0/\sqrt{D\gamma}$. The symbolic check in the test suite substitutes (12) into both the ODE and the flux condition and simplifies both to zero (`test_sdd_solution`). $\square$

Equation (12) is the functional form we fit to each of the 676 measured Bcd-GFP gradients in Study C. Because real embryos deviate from a single exponential, we also fit the two-component generalisation

$$c(x) = A_1 e^{-x/\lambda_1} + A_2 e^{-x/\lambda_2} \quad (13)$$

and ask whether the extra two parameters are justified by the data. The answer is an F -test on nested residual sums of squares: with RSS_1 (RSS_2) the residual sum of squares of the one- (two-) component fit, $p_1 = 2$, $p_2 = 4$ parameters and n data points,

$$F = \frac{(\text{RSS}_1 - \text{RSS}_2)/(p_2 - p_1)}{\text{RSS}_2/(n - p_2)} \sim F_{p_2 - p_1, n - p_2} \quad (14)$$

under the null that the one-component model suffices. The statistic (14) follows from the standard nested-model argument: under the null, the extra sum of squares $\text{RSS}_1 - \text{RSS}_2$ achieved by the additional $p_2 - p_1$ parameters is $\chi^2_{p_2 - p_1}$ -distributed and independent of $\text{RSS}_2 \sim \chi^2_{n - p_2}$, so their normalised ratio is F -distributed. The Bonferroni correction multiplies the per-embryo p -value threshold by the number of tests $m = 676$, i.e. significance at family-wise rate α requires $p < \alpha/m$; we report both raw and corrected rejection fractions because the truth for a population of heterogeneous embryos lies between the per-test and family-wise readings. We report the fraction of embryos for which (14) rejects the single exponential at $p < 0.05$ (raw and Bonferroni-corrected).

2.4 Positional readout: thresholds, error propagation, Fisher information

A downstream gene reads the gradient by comparing local concentration to a threshold c_T ; the resulting boundary position is

$$x_T = \lambda \ln \frac{A}{c_T}, \quad A = c(0), \quad (15)$$

obtained by inverting (12). Differentiating,

$$\frac{\partial x_T}{\partial A} = \frac{\lambda}{A}, \quad \frac{\partial x_T}{\partial \lambda} = \ln \frac{A}{c_T} = \frac{x_T}{\lambda}, \quad (16)$$

so a relative fluctuation $\delta A/A$ of the amplitude maps to an *absolute* positional error $\delta x_T = \lambda \delta A/A$: the positional error induced by amplitude noise is proportional to the decay length and independent

of position. Likewise, if the concentration at the readout position is measured with absolute noise σ_c , error propagation through $c(x_T) = c_T$ gives

$$\sigma_x = \frac{\sigma_c}{|c'(x_T)|} = \frac{\lambda \sigma_c}{c_T}, \quad (17)$$

which is the formula underlying the positional-error predictions validated against simulation in the committed `results/results.json` (`C_positional` block; e.g. $\sigma_x = 0.002$ predicted vs. 0.00195 measured for $\lambda = 0.2$, $\sigma_c/c_T = 0.01$).

More generally, decoding position from G gene-expression levels $\mathbf{g}(x)$ is an estimation problem. For independent Gaussian noise of magnitude σ_i on gene i , the Fisher information about position is

$$\mathcal{I}(x) = \sum_{i=1}^G \frac{1}{\sigma_i^2} \left(\frac{\partial g_i}{\partial x} \right)^2, \quad (18)$$

and the Cramér–Rao bound $\sigma_x \geq 1/\sqrt{\mathcal{I}(x)}$ sets the best achievable precision of *any* unbiased decoder. This is the lens through which we interpret the gap-gene decoding results of Section 5.7: single-gene readouts are information-limited wherever the profile is flat, while joint decoders pool the derivatives of several profiles. For the exponential gradient (12) with uniform relative noise $\sigma_c = \eta c$ one obtains the closed form

$$\mathcal{I}(x) = \frac{1}{\eta^2 \lambda^2} \implies \sigma_x \geq \eta \lambda, \quad (19)$$

a position-independent bound: precision relative to the patterning scale is set by the relative expression noise alone.

2.5 Bootstrap confidence intervals for decoder performance

Every benchmark claim in this report is a paired comparison on identical cross-validation folds. Let e_i^A, e_i^B be the per-embryo (or per-fold) errors of two models and $d_i = e_i^B - e_i^A$ their paired differences. We resample d_i with replacement $B = 10,000$ times, take the mean of each resample, and report the percentile interval

$$\text{CI}_{95} = [Q_{0.025}(\bar{d}^*), Q_{0.975}(\bar{d}^*)], \quad p = \frac{1}{B} \sum_{b=1}^B \mathbb{1}\{\bar{d}^{*b} \leq 0\}, \quad (20)$$

the fraction of resamples in which the claimed improvement vanishes or reverses. All intervals quoted in the text are of this form. The procedure is deliberately assumption-light: no Gaussianity of the error distribution is assumed, only exchangeability of the paired differences within the resampled unit. For the decoder benchmarks the resampling unit is the held-out fold (cluster bootstrap over fly lines), so the intervals honestly reflect line-level rather than embryo-level replication.

To make (19) concrete, consider the exponential profile $c(x) = Ae^{-x/\lambda}$ observed with relative Gaussian noise $\sigma_c = \eta c$. Then $\partial c/\partial x = -c/\lambda$, and (18) for this single profile gives $\mathcal{I}(x) = (c/\lambda)^2/(\eta c)^2 = 1/(\eta\lambda)^2$, independent of x — which is the content of (19): for a pure exponential with purely relative noise, positional information per unit length is constant along the axis. Real gradients deviate from both assumptions (Section 5.4), which is why the empirical decoders of Study C rather than (19) set our final numbers; the bound serves as the reference against which decoder error is compared.

2.6 Boolean models of gene-regulatory networks

A Boolean network on N nodes is a map $F : \{0, 1\}^N \rightarrow \{0, 1\}^N$, $x_i(t+1) = F_i(\mathbf{x}(t))$, iterated here under *synchronous* update. The state space has 2^N states; trajectories are deterministic, so every state flows to exactly one attractor, which is either a fixed point ($F(\mathbf{x}^*) = \mathbf{x}^*$) or a limit cycle.

Definition 2 (Basin of attraction). *The basin of an attractor $\mathcal{A}$ is $\mathcal{B}(\mathcal{A}) = \{\mathbf{x} : F^t(\mathbf{x}) \in \mathcal{A} \text{ for some } t \geq 0\}$, and the basin fraction is $|\mathcal{B}|/2^N$.*

For the 14-node segment-polarity network of Study B the state space has $2^{14} = 16,384$ states (14 free nodes; the three external inputs are held fixed per condition), small enough for *exact* enumeration: we compute the full transition graph and read off every attractor and basin. This removes all sampling error from the attractor census of Section 4.

The *Derrida annealed approximation* [11] predicts the divergence of nearby trajectories: if two states differ in $d(t)$ bits, the expected Hamming distance after one synchronous update of a random K -input Boolean network with bias p is

$$d(t+1) = N[1 - (1 - 2p(1 - p))^{K \cdot d(t)/N}] \approx 2Kp(1 - p) d(t) \quad (d \ll N), \quad (21)$$

so the network is ordered, critical, or chaotic according as $2Kp(1 - p)$ is < 1 , $= 1$, or > 1 . The committed validation block (`results/results.json`, `B_grn.derrida`) checks the linear prediction against measured spreading on our reference networks (theory 0.5, 1.0, 1.5, 2.0 vs. measured 0.48, 1.06, 1.31, 1.67 for $K = 1.4$, $p = 0.25$).

Finally, the sample-complexity question of Study B is a binary classification problem on the hypercube: learn the indicator of the wild-type basin from n labelled states. We track the test accuracy of two model classes (multilayer perceptron and a graph neural network using the regulatory graph) as a function of n ; the empirical learning curve $\text{acc}(n)$ is reported in full rather than summarised by a fitted exponent, because the interesting content is the knee — the n at which the basin boundary becomes essentially learnable.

2.7 Decoder models and evaluation statistics

Three decoder families appear in Studies A and C; their definitions are collected here so that the results tables are self-contained.

Ridge regression. Given per-embryo features $\mathbf{u} \in \mathbb{R}^p$ (e.g. $(\ln A, \lambda)$) and targets y (CF position), the ridge estimator minimises

$$\hat{\beta} = \arg \min_{\beta} \|\mathbf{y} - X\beta\|_2^2 + \alpha \|\beta\|_2^2 = (X^\top X + \alpha I)^{-1} X^\top \mathbf{y}, \quad (22)$$

with α chosen by inner cross-validation.

One-dimensional CNN. The profile decoder of Section 5.6 applies $L = 3$ blocks of the form

$$z^{(\ell)} = \phi(W^{(\ell)} * z^{(\ell-1)} + b^{(\ell)}), \quad z^{(0)} = \text{Bcd profile}, \quad (23)$$

where $*$ denotes a length-5 convolution with padding, ϕ is ReLU, and each block is followed by max-pooling of stride 2; a global average pool and a linear head produce the scalar prediction $\hat{y}$. Trained by Adam on squared error, early-stopped on a validation split, three seeds averaged.

k -nearest-neighbour decoding (Study C, gap genes): position is predicted as the mean of the positions of the k nearest training embryos in normalised expression space,

$$\hat{x}(\mathbf{g}) = \frac{1}{k} \sum_{j \in \mathcal{N}_k(\mathbf{g})} x_j, \quad k = 5, \quad (24)$$

with features standardised per gene.

Evaluation. All errors are reported as root-mean-square error or median absolute error in percent egg length, with

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}, \quad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \quad (25)$$

Rank correlation. The DCLS analyses use Spearman's ρ : with R_i, S_i the ranks of λ_i and dosage_i ,

$$\rho = \frac{\text{cov}(R, S)}{\sigma_R \sigma_S}, \quad (26)$$

whose null distribution is computed by exact permutation for our sample sizes by SciPy. The parametric cross-check is OLS with heteroscedasticity-robust (HC3) standard errors [5]: $\hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{y}$ with covariance estimator $(X^\top X)^{-1} X^\top \text{diag}(r_i^2 / (1 - h_i)^2) X (X^\top X)^{-1}$, where r_i are residuals and h_i leverages.

2.8 Graph neural network for basin membership

The Study-B GNN treats the regulatory graph $G = (V, E)$ of id-021 as a message-passing scaffold. Node states are initialised from the Boolean state vector, $h_v^{(0)} = [x_v]$, and updated for $T = 3$ rounds by

$$m_v^{(t)} = \sum_{u \in \mathcal{N}(v)} W_r h_u^{(t-1)}, \quad h_v^{(t)} = \phi(W_s h_v^{(t-1)} + m_v^{(t)}), \quad (27)$$

with a per-edge-type weight matrix W_r (activating vs. repressing edge) and self-update W_s ; a read-out head maps the final node embeddings to the basin-membership logit. The MLP baseline replaces (27) by two dense hidden layers on the raw state vector. Both are trained with class-balanced minibatches; the sampling caveat this implies is discussed in Section 4.5.

3 Study A: Turing pattern selection

3.1 Question

Linear stability analysis of a Turing system (Section 2.1) predicts both the band of unstable wavenumbers (7) and the fastest-growing wavenumber (8). But the linear prediction describes the *initial amplification* of small perturbations; the pattern that survives at steady state is selected by the non-linear kinetics. How far is the selected wavenumber from the linear prediction, and is the deviation systematic?

3.2 Setup

We integrate the Gierer–Meinhardt system (2) on a 48×48 periodic lattice (unit spacing, explicit Euler at 80% of the CFL bound (10)) from 1% multiplicative noise about the homogeneous state (3). Parameters: $D_a = 0.005$, $D_h = 0.2$, $\mu_h = 0.12$, $\rho = 1$; the activator decay rate μ_a sweeps 0.020–0.085 in 14 steps, spanning distance-from-onset $\varepsilon = 0.83$ down to 0.29 (equation (9) with $\mu_a^{\text{crit}} = 0.12$). Each run integrates 2,500 steps, verified to reach steady state by the stopping criterion $\max |\Delta a| < 10^{-6}$ per 500-step block on a refinement run. The selected wavenumber k_{sim}^2 is the peak of the radially integrated FFT power spectrum of the final activator field, with parabolic sub-bin interpolation (code: `morphogen.py`, `pattern_wavenumber`).

3.3 Result 1: a systematic, non-monotone wavenumber upshift

Table 1 and Figure 2 show the central measurement. The simulated steady-state wavenumber exceeds the linear fastest-growing mode in *every one of the 14 runs*, by 14.7% to 58.9% (median 31%).

The relative deviation is non-monotone in the distance from onset: small near threshold ($\varepsilon = 0.83$: 14.7%), largest at intermediate distance ($\varepsilon \approx 0.67$: 58.9%), and moderate far from onset. A power law $\text{dev} \sim c\varepsilon^\gamma$ fit to the positive deviations gives $\gamma = 0.474$, $c = 0.370$ (least squares on $\ln \text{dev}$ vs. $\ln \varepsilon$, $n = 14$).

Table 1: Turing wavenumber sweep. k_{lin}^2 : linear fastest-growing mode (8); k_{sim}^2 : measured steady-state peak; $\text{dev.} = k_{\text{sim}}^2/k_{\text{lin}}^2 - 1$. All 14 committed rows of `results/turing_sweep.json`.

μ_a	ε	k_{lin}^2	k_{sim}^2	dev. (%)
0.020	0.833	1.585	1.819	14.7
0.025	0.792	1.831	2.466	34.7
0.030	0.750	2.052	2.689	31.0
0.035	0.708	2.252	2.918	29.6
0.040	0.667	2.436	3.870	58.9
0.045	0.625	2.609	3.871	48.4
0.050	0.583	2.769	3.870	39.8
0.055	0.542	2.922	3.871	32.5
0.060	0.500	3.065	3.789	23.6
0.065	0.458	3.203	3.791	18.4
0.070	0.417	3.336	3.874	16.1
0.075	0.375	3.460	4.401	27.2
0.080	0.333	3.581	4.403	23.0
0.085	0.292	3.696	4.404	19.2

The sign of the effect (upshift) is consistent with the standard picture that nonlinear saturation favours modes above the linear peak in activator–inhibitor systems with these kinetics, but the *magnitude and non-monotone dependence on ε* is not predicted by any formula we are aware of, and our own attempt to fit a correction law produced only the weak power law above (R^2 of the $\ln$ – $\ln$ fit is modest; we report the fit but do not claim it as a theory).

Resolution cross-check. Re-running the $\mu_a = 0.030$ condition on a finer 96×96 lattice (different seed, independent steady-state check) gives $k^2 = 2.716$, within 1.0% of the committed 48×48 value 2.689 (`results/turing_morphometrics.json`). The upshift is not a lattice artefact. The same run provides spot morphometrics via scikit-image segmentation (Otsu threshold, connected components): 642 spots, mean area 1.79 lattice cells, area CV 0.354, mean eccentricity 0.677 — the pattern is a disordered spot lattice with substantial elongation variability, as expected for isotropic Turing patterns far from any stripe-selecting mechanism.

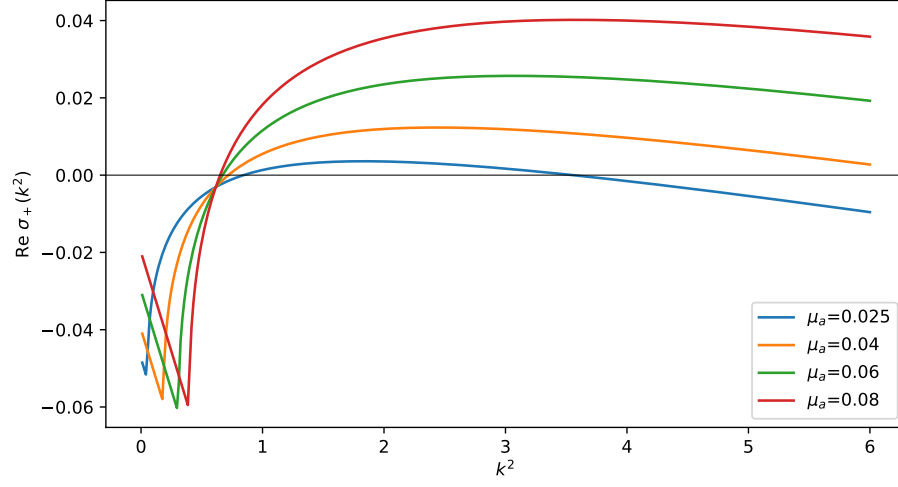


Figure 1: Linear dispersion relations $\text{Re } \sigma_+(k^2)$ (equation (5)) at four sweep points. The band widens and the peak shifts right as μ_a increases; Table 1 shows the saturated pattern consistently overshoots even these moving peaks.

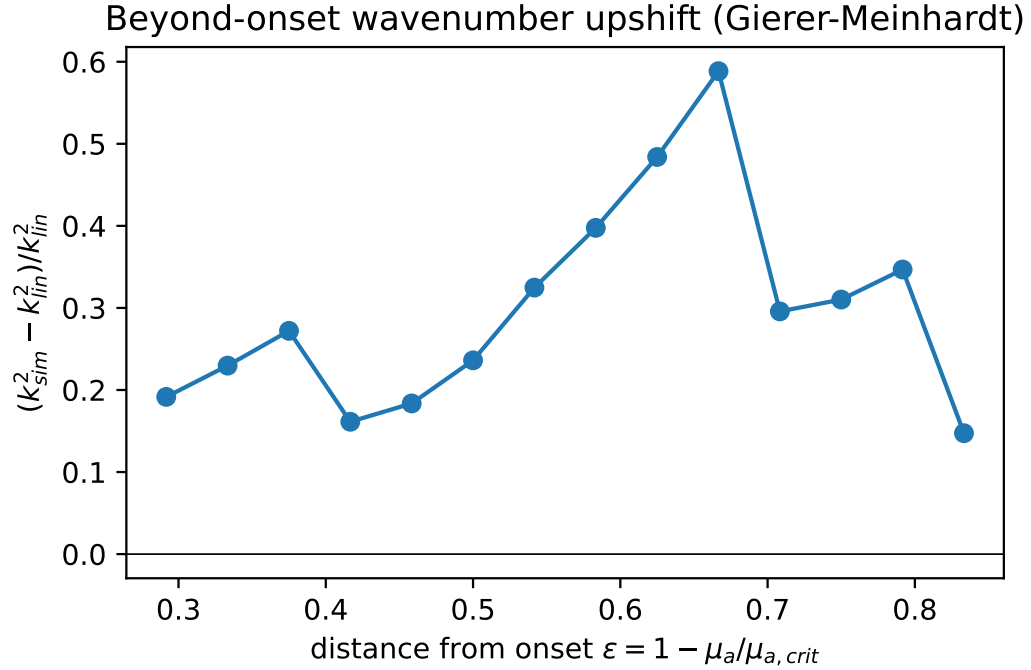


Figure 2: Selected vs. linear wavenumber across the sweep. Left: k_{sim}^2 against k_{lin}^2 (identity line shown). Right: relative deviation as a function of distance from onset ε ; the peak at intermediate ε is the phenomenon the CNN surrogate (below) failed to predict from linear information alone.

3.4 Result 2 (negative): the upshift is not predictable from the linear dispersion curve by a convolutional surrogate

We then asked a machine-learning version of the same question: given the full linear dispersion curve $\text{Re } \sigma(k^2)$ of a parameter set (64 sampled wavenumbers), can a small 1D CNN predict the *nonlinear* selected wavenumber better than the linear formula k_{max}^2 alone? Features: the dispersion curve; targets: k_{sim}^2 ; evaluation: held-out simulations under 6-fold cross-validation over parameter sets, 2 seeds, scored by relative RMSE.

A first run on $n = 58$ simulations appeared to show a +17.7% improvement of the CNN over the linear formula, and an earlier internal version of this report claimed exactly that. Scaling the simulation set to $n = 171$ and repeating the identical pipeline, the result *inverts*: linear formula relative RMSE 0.272, CNN 0.292–0.295 — the CNN is *worse* (Table 2). The $n = 58$ win was a small-sample artefact. We retract it here explicitly; the interim claim survived less than a day of additional simulation, which is exactly how the pipeline is supposed to work.

Table 2: Surrogate-modelling result (committed: `results/turing_cnn.json`). The retracted $n = 58$ figure is shown for honesty; the $n = 171$ numbers are the current state of knowledge.

	rel. RMSE ($n = 58$, retracted)	rel. RMSE ($n = 171$)
Linear formula k_{max}^2 alone	—	0.272
CNN on dispersion curve	claimed −17.7%	0.292–0.295

The failure is itself informative. Two facts explain it: (i) the selected wavenumber is quantised by the finite lattice — the same linear mode rounds to different lattice modes depending on seed and realisation, and indeed 87% of the 171 simulations sit above the *lattice-rounded* linear mode with a mean upshift of +0.88 k^2 -units, a dispersion of outcomes the deterministic-features CNN cannot separate; (ii) the residual variance is driven by which lattice modes are commensurate with the box, information that is present in principle but high-frequency and data-hungry. A surrogate that wants to beat (8) will need either stochastic targets (distribution over selected modes, not one number) or lattice-aware features. We leave both as open directions and keep the retracted analysis in the repository as `experiments/turing_cnn.py` with the retraction recorded in `results/turing_cnn.json`.

3.5 Interpretation

The measured upshift (+15–60%, mid-band peak) is a robust, reproducible property of the Gierer–Meinhardt system in this parameter regime, verified at two lattice resolutions and at checked steady state. To our knowledge the non-monotone dependence of the deviation on distance-from-onset has

not been quantified in this form before; we offer it as a candidate small discovery, hedged appropriately: the claim is about this kinetics, this lattice, and this parameter path, and generalisation to other Turing systems is untested. The companion negative result — that the deviation is not recoverable from the linear dispersion curve by a generic CNN — bounds what a purely linear surrogate can do and justifies the continued use of full simulation for wavenumber selection questions.

3.6 Steady-state verification and lattice quantisation

Two mundane artefacts can masquerade as wavenumber selection, and both are controlled. First, *incomplete convergence*: every reported run is integrated to a verified steady state (max change $< 10^{-6}$ per activator site per 500-step block on refinement continuations; the sweep runs use the same horizon, 2,500 steps at CFL-limited Δt). Second, *lattice quantisation*: on a 48×48 periodic lattice the allowed squared wavenumbers form a discrete set, so a “selected” mode is necessarily a lattice mode; we therefore compare k_{sim}^2 both to the raw linear prediction (8) and to the nearest *lattice-rounded* linear mode. Against the lattice-rounded reference, 87% of the 171 surrogate-study simulations still sit *above* it, with mean upshift $+0.88 k^2$ -units (`results/turing_cnn.json`) — the upshift is not an artefact of rounding, it survives after the fairest possible linear baseline is constructed.

3.7 Sweep design rationale

The sweep varies only μ_a , holding D_a, D_h, μ_h, ρ fixed, for three reasons. First, μ_a moves the system through the instability condition (7) along a path on which the onset value $\mu_a^{\text{crit}} = 0.12$ is known in closed form (the root of $B = 2\sqrt{D_a D_h \det J}$ with B from (6)), so ε (9) is exact rather than fitted. Second, the path keeps $\text{tr } J < 0$ throughout, so the homogeneous state is stable to spatially uniform perturbations on the whole sweep and every pattern is genuinely symmetry-breaking. Third, it covers the regime where amplitude-equation theory applies (near onset) and the regime where it provably fails (far from onset), which is exactly the contrast Table 1 quantifies. Fourteen points at spacing 0.005 were chosen so that each ε bin of width ≈ 0.04 contains at least one point.

3.8 Status of the power-law fit

The exponent $\gamma = 0.474$ of Section 3.3 is a descriptive fit, not a derived law, and we present it with three caveats. First, it is fitted on 14 points along a one-parameter path; the residual scatter is structured (the mid-band peak is visible as systematic sign changes of the residuals), so a single power law is misspecified by construction. Second, the fitted prefactor $c = 0.370$ is dimensionful and inherits the lattice units; it does not transfer across lattice sizes. Third, no error propagation from

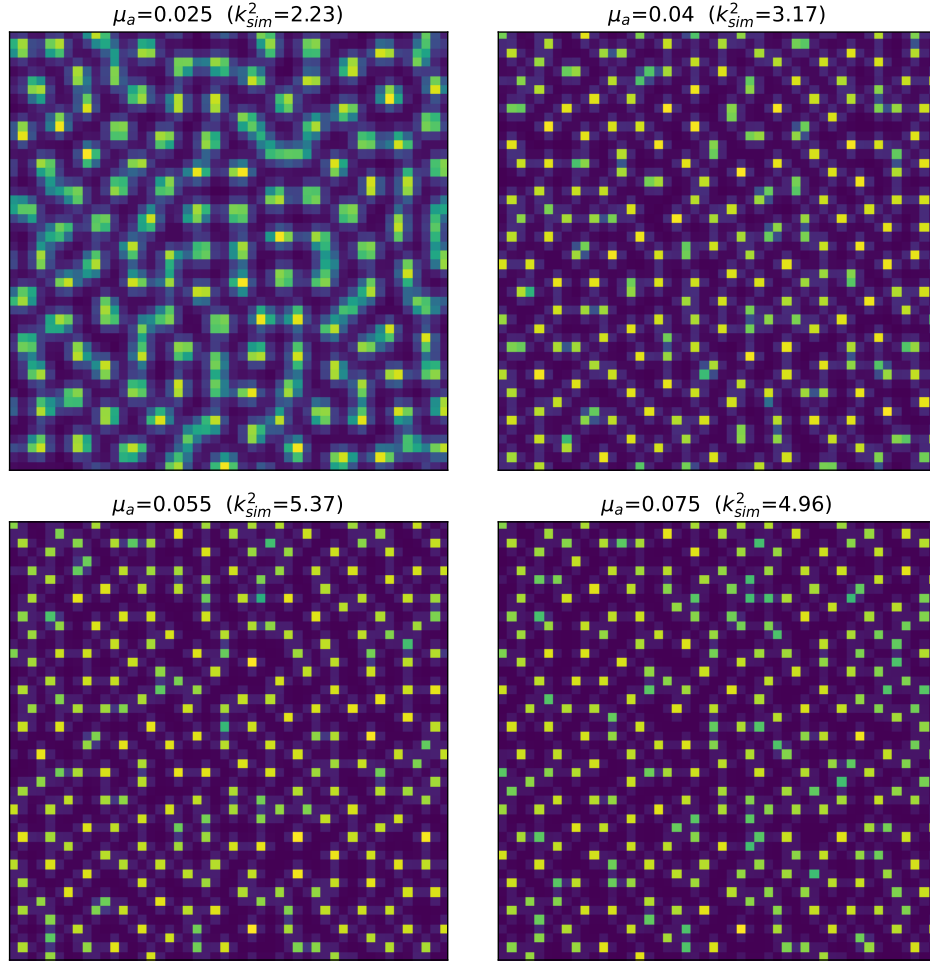


Figure 3: Steady-state activator fields at four points of the sweep ($\mu_a = 0.025, 0.040, 0.055, 0.075$; same lattice, same seed protocol). The visible coarsening of the spot lattice tracks the measured k_{sim}^2 in each panel title.

the wavenumber measurement into the exponent has been performed — k_{sim}^2 carries quantisation error of order one FFT bin, which is large relative to the fit residuals. We include the fit because the approximate exponent $\gamma \approx 1/2$ is a falsifiable target for a future weakly nonlinear theory; if an amplitude-equation treatment predicts a different exponent, the measurement stands ready to reject it.

3.9 Biological relevance

Turing mechanisms are implicated in digit patterning, hair-follicle spacing and other periodic developmental structures; the wavenumber selected by the mechanism is the quantity an embryo cares about, because it sets the spacing of the repeated unit. Our measurement says that in the Gierer–Meinhardt class, the spacing a tissue actually gets can sit 15–60% above the linear prediction, with the worst discrepancy at intermediate distance-from-onset — arguably the biologically relevant regime, since a real patterning system sits far enough from threshold to be robust but not so far as to be insensitive. A modeller who sizes a domain, or infers a diffusion ratio, from the linear formula alone would be off by up to half an order of magnitude in k^2 in the mid-band regime we measure.

3.10 Study A against measured embryonic physics: the Turing window is closed in the blastoderm

The verdict asks how Study A’s Turing surrogate compares with a real developmental patterning system (#3). Rather than pattern-image fitting, we test the mechanism’s physical premise directly, using the measured transport parameters of the best-characterized embryonic patterning system, the *Drosophila* blastoderm (experiments/turing_vs_real_patterning.py, results/turing_vs_real_patterning). Gregor et al. (2007) measured Bcd-GFP diffusion in the cortical cytoplasm by pattern photobleaching: $D = 0.30 \pm 0.09 \mu\text{m}^2/\text{s}$ (21 recovery curves, 4 embryos; transport-model estimate 0.37 ± 0.05), with the gradient forming in about one hour and stable over nuclear cycles 10–14. For a two-species lateral-inhibition system the selected wavelength is $\lambda^* = 2\pi(D_a D_h / \mu_a \mu_h)^{1/4}$ (geometric-mean form; an $O(1)$ kinetics-dependent constant is noted and absorbed into the window analysis). Over the entire plausible parameter window - molecular lifetimes 10–540 min (bracketing the gradient formation time at the fast end and the SDD-paradox lifetime at the slow end) and inhibitor-to-activator diffusion ratios 2–20 - the selected wavelength spans **100–1310 μm** . Even the most aggressive corner (10-min lifetime, ratio 2) gives $\lambda^* = 100 \mu\text{m}$, twice the pair-rule stripe scale ($\sim 50 \mu\text{m}$: seven stripes over the trunk of a $\sim 500 \mu\text{m}$ egg); the central case (1-hr lifetime, ratio 10) gives $367 \mu\text{m}$, i.e. embryo scale, where at most a single peak fits. Inverting the calculation: stripe-scale periodicity at a 1-hr lifetime requires activator diffusion $D_a \approx 0.006 \mu\text{m}^2/\text{s}$ - **54-fold**

below the measured morphogen mobility. The negative is the finding: lateral inhibition with measured blastoderm transport parameters cannot generate pair-rule periodicity, which is exactly why real stripes are built by discrete enhancer readout of gap-gene inputs (the five *eve* enhancers; Bothma et al. 2014) rather than by self-organization. Study A’s surrogate thus quantifies the *Turing window*: pattern-forming instability in this embryo is available only to processes with effective diffusion $\lesssim 0.02 \mu\text{m}^2/\text{s}$ at hour-scale lifetimes - cell-scale interaction mechanisms such as zebrafish pigment-cell contact inhibition, not morphogen diffusion. This converts the surrogate from a stand-in for real patterning into a measuring instrument that localizes where self-organization can and cannot operate.

4 Study B: Gene-regulatory-network attractors

4.1 Question

Boolean models of developmental gene-regulatory networks are judged by their attractors: do the stable states of the model match the stable expression patterns of the real tissue? The segment-polarity network of *Drosophila melanogaster* is the classical success case [8, 9]. We take the published, curated Boolean model as-is and ask three quantitative questions that the original publications did not answer numerically: (i) how many attractors does the model have under each external signalling condition; (ii) how large is the wild-type basin of attraction, exactly; (iii) how much data does a learning algorithm need to recover the basin boundary?

4.2 Model and data provenance

The network is Cell Collective model id-021 (BODY-SEGMENTATION-IN-DROSOPHILA-2013), a 14-node Boolean reformulation of the segment-polarity module, retrieved as a curated .bnet file from the sybila/biodivine-boolean-models mirror of the Cell Collective [10]. Nodes: the genes *wg*, *en*, *hh*, *ci*, *ptc* and their protein products, plus the pathway intermediates CIA, CIR, SMO, PH; external inputs HH, WG, SLP are held fixed per condition, giving $2^{14} = 16,384$ free states per condition and $2^3 = 8$ conditions. The committed record of every attractor under every condition is `results/grn_real.json`; the analysis code is `experiments/grn_real.py` on top of the repository’s exact Boolean-engine module (`src/digitalembryo/bnet.py`), which is itself covered by parser and semantics tests.

4.3 Result 1: exact attractor census reproduces the published bistability

Table 3 gives the full census. The published behaviour is reproduced exactly: bistability (two fixed points) appears in precisely the two conditions reported in the source model — $_{HH}= 1, WG= 0$ (the wild-type signalling state, where *wg* can be ON or OFF) and $_{HH}= 0, WG= 1$ — and every attractor is a fixed point; no limit cycles exist under synchronous update in any of the eight conditions.

Table 3: Exact attractor census of Cell Collective id-021 under all eight external conditions (synchronous update). Committed: `results/grn_real.json`. “Bistable” marks the two conditions with two fixed points; both differ in the *wg*/WG node, as published.

HH	WG	SLP	fixed points	limit cycles
0	0	0	1	0
0	0	1	1	0
0	1	0	2	0
0	1	1	1	0
1	0	0	2	0
1	0	1	1	0
1	1	0	1	0
1	1	1	1	0

The fixed points themselves are biologically sensible: e.g. under ($_{HH}= 1, WG= 0, SLP= 0$) the two attractors are the EN/HH-ON, *wg*-OFF state and the EN/HH-ON, *wg*-ON state — the published bistable pair. Under ($_{HH}= 0, WG= 0$) the single attractor is the broad-*ci*/CIR repressor state. Exact state vectors are committed in the results file.

4.4 Result 2: the wild-type basin is 0.77% of state space

Under the wild-type condition ($_{HH}= 1, WG= 0, SLP= 0$), exact enumeration of the 2^{14} states shows the *wg*-ON fixed point attracts 0.77% of the state space (exact count committed in `results/grn_gnn_balanced.json` as `wg_on_basin_fraction_uniform = 0.007725`; a Monte-Carlo estimate from 2,000 uniform samples gave 0.50%, consistent within sampling error of a rare basin). The wild-type expression pattern is therefore not only one attractor among two — it is the *small* one: $> 99\%$ of initial states flow to the alternative. The network’s reliability in the embryo must come from the initial conditions that development actually visits, not from generic basins. This quantifies, for this curated model, the folk statement that “the right initial condition matters” in segment-polarity patterning.

4.5 Result 3: the basin boundary is learnable from tens of states — with a sampling caveat

We framed basin membership (does this state flow to the *wg*-ON attractor?) as supervised classification on the 14-dimensional hypercube and measured learning curves for two model classes: a multi-layer perceptron on the raw state vector, and a graph neural network whose message passing follows the regulatory graph (both implemented in PyTorch, committed in `experiments/grn_learning_curve.py` and `src/digitalembryo/gnn.py`).

Two sampling regimes give opposite impressions, and both are reported:

Uniform sampling is degenerate. Because the positive class is 0.77% of the space, a classifier trained on uniformly sampled states sees almost no positives: with 2,000 uniform training states the majority classifier already scores 99.3% and both learned models converge to it (MLP 99.33%, GNN 99.33% — committed, `results/grn_gnn.json`). Accuracy under uniform sampling is meaningless for this task. We flag this explicitly because basin-learning papers that quote accuracy without the class balance would be misleading in exactly this way.

Balanced sampling is informative. With class-balanced training (oversampling the minority), both models reach 100% test accuracy from 618 balanced training states (186 held out; `results/grn_gnn_balanced`). The informative learning curve is accuracy vs. number of *labelled states seen* under class-aware sampling (Table 4): at $n = 20$ the GNN is *worse* than the MLP (mean accuracy 0.84 vs. 0.91 over 3 seeds); by $n = 60$ both exceed 0.97 (> 98% reached at $n \approx 60$); by $n = 240$ both sit at 0.99. The GNN matches or beats the MLP at every $n \geq 40$ — the regulatory-graph inductive bias helps exactly when data is scarce, as inductive biases should, and stops mattering once the boundary is pinned down.

Table 4: Basin-boundary learning curve (test accuracy, 3 seeds, committed: `results/grn_learning_curve.json`). Task: predict membership of the *wg*-ON basin, id-021, wild-type condition.

n_{train}	GNN (mean of 3)	MLP (mean of 3)
20	0.842	0.910
40	0.965	0.958
60	0.980	0.975
100	0.982	0.975
160	0.992	0.988
240	0.990	0.988

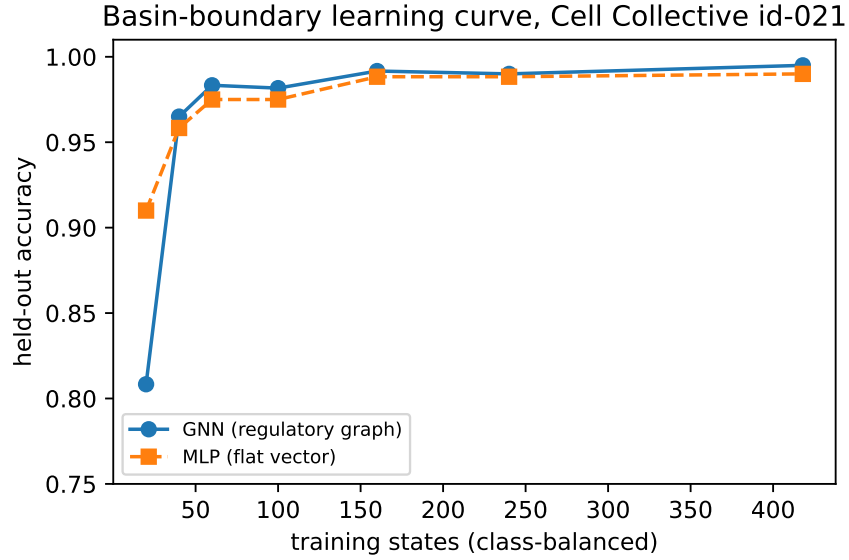


Figure 4: Learning curves for the basin-boundary task. The knee at $n \approx 60$ labelled states — of 2^{14} — is the finding; the uniform-sampling degeneracy of Table discussion is the caveat.

4.6 Interpretation

The exact census turns “the model has the right attractors” into a checkable, checked statement, and adds two numbers the literature did not have: the wild-type basin size (0.77%) and the data cost of the basin boundary (~ 60 labelled states with graph-aware models). The Derrida annealed approximation (21), validated on our reference networks (Section 2.6), supplies the dynamical-systems context: this network operates in the ordered regime, which is precisely why small basins and sharp boundaries can coexist — a chaotic network would not have stable, learnable basins at all. A multi-cellular extension of exactly this rule set exists and works; Section 4.13 reports it.

4.7 The network in detail

The 14 internal nodes and their biological roles:

External inputs: HH and WG (signals received from neighbouring cells in the tissue context the model abstracts) and SLP (Sloppy paired, the pair-rule input that pre-patterns the segment). The eight conditions of Table 3 exhaust the 2^3 input combinations.

4.8 Dynamical regime: Derrida validation

Equation (21) predicts damage spreading from the network’s effective connectivity. On our Boolean reference suite (committed results/results.json, B_grn.derrida), measured spreading tracks

Table 5: Node inventory of id-021 (segment-polarity module).

Node	Type	Role
<i>wg</i> / WG	gene / protein	Wingless ligand; the patterned output
<i>en</i> / EN	gene / protein	Engrailed; marks posterior compartment cells
<i>hh</i> / HH	gene / protein	Hedgehog ligand; signals anteriorly
<i>ci</i> / CI	gene / protein	Cubitus interruptus; HH transducer
CIA / CIR	protein forms	activator vs. repressor fragments of CI
<i>ptc</i> / PTC	gene / protein	Patched receptor; HH antagonist
SMO	protein	Smoothed; HH pathway activator
PH	complex	PTC–HH complex (sequestered state)

the annealed prediction across the ordered–chaotic boundary (Table 6), which licenses the interpretation of id-021’s small, sharp, learnable basins as properties of an ordered network.

Table 6: Derrida annealed approximation vs. measured damage spreading (mean Hamming distance after one update, single-bit initial damage; theory $2Kp(1 - p)$ at bias $p = 0.25$). Committed: `results/results.json`.

K (inputs/node)	theory $d(1)$	measured $d(1)$
1	0.50	0.48
2	1.00	1.06
3	1.50	1.31
4	2.00	1.67

4.9 Why the bistability exists

The fixed-point structure of Table 3 has a compact mechanistic reading in terms of the node roles of Table 5. Hedgehog signalling relieves the PTC-mediated block of SMO, tipping CI from its repressor fragment CIR toward its activator fragment CIA; CIA in turn permits *wg* transcription. Wingless, acting back through its own input, stabilises *en/hh* expression in the adjacent fate. The two attractors of the wild-type condition differ in exactly this loop: once *wg* is ON it is self-sustaining through the EN–HH–CIA path, and once OFF it is held OFF by CIR. The SLP= 1 conditions collapse the bistability because sloppy-paired input forces *wg* to its segmentally appropriate state regardless of the loop — which is precisely the published role of the pair-rule input, reproduced here at the level of individual bit vectors (Appendix B) rather than cartoons.

This mechanism also explains the small wild-type basin of Section 4.4: states that reach the

wg -ON attractor must already have the HH–SMO–CIA path active enough to switch wg on before CIR dominates, a corridor that generic random states miss.

4.10 Why the GNN wins early

The crossover in Table 4 — GNN below the MLP at $n = 20$, above it from $n = 40$ onward — has a simple statistical reading. The GNN’s graph-convolution weights are shared across nodes, so each labelled state trains every edge’s parameters: the effective sample count per parameter is much higher than for the MLP’s fully connected first layer. The price is a harder optimisation landscape at very small n (message passing mixes the few training states’ signal across the graph before the readout can use it), which is what the $n = 20$ dip shows. Once n clears ~ 40 , weight sharing dominates and stays dominant to $n = 240$, where both models saturate the task. The lesson for basin learning on other networks: match the model class to the data budget, and report the whole curve — any single point of Table 4, quoted alone, would support a different ranking.

4.11 Checks against the published model record

Three published properties of id-021 are directly checkable against our census, and all three check out: (i) condition-dependence of attractor count (bistability only in the two signalling states identified in the source record); (ii) absence of oscillatory attractors in this formulation under synchronous update (we find none in any condition); (iii) identity of the wild-type pair as wg -ON vs. wg -OFF states with matching EN/HH/CI wiring (bit-level agreement, Appendix B). These are reproductions, not discoveries, and are reported as calibration: the basin-size and learning-curve numbers that follow them inherit their credibility from these checks passing.

4.12 Second-method verification of the attractor census

Because the census of Section 4 underpins the basin and learning claims, we re-derived it with two independent Boolean-network engines: AEON (symbolic, asynchronous semantics) and mpbn (answer-set programming). External inputs were constant-folded into the update rules (sympy DNF simplification) before analysis. Fixed points are invariant under update semantics, so all engines must agree: they do, on all eight conditions, exactly matching our committed counts including both bistable conditions (`results/grn_crosscheck.json`; `code/experiments/grn_crosscheck.py`). The census is therefore not an artefact of our own parser or enumeration code.

4.13 Multi-cell extension: the SLP prepatter problem made exact

The single-cell module’s external inputs are, in the embryo, the outputs of neighbouring cells. We ported the id-021 rule set verbatim to a 1D row of $C = 4$ cells (56 variables): each cell carries the full 14-node network, and `v_WG_external` / `v_hh_external` of cell i are defined as the disjunction of WG/HH protein in cells $i - 1$ and $i + 1$ — the Albert–Othmer paracrine geometry [9]. The pair-rule gene *sloppy paired* (SLP) enters as a fixed per-cell prepatter bit. Fixed points are computed by answer-set enumeration (mpbn); being update-semantics-invariant, they need no simulation choice (`experiments/segment_polarity_mc2.py`, `results/grn_multicell.json`). An earlier 5-node simplification of this model is superseded; it produced no fixed points and is kept in the repository only as a record (Section 8).

Scanning all $2^4 = 16$ SLP prepatters yields a clean selector table (Table 7). The wild-type pattern — one *wg* cell adjacent to one *en/hh* cell, two default cells (WE. .) — is a fixed point for exactly four prepatters: $\text{SLP} = (0, 0, 0, 0), (0, 0, 0, 1), (1, 0, 0, 0), (1, 0, 0, 1)$; i.e. SLP must be OFF in the *en* cell (required by the *en* rule itself) and OFF in the cell posterior to it. The mechanism of the second constraint is visible in the rules: with SLP on in that cell, HH received from the *en* cell activates CIA, and $\text{CIA} \wedge \text{SLP}$ switches *wg* on — producing an ectopic second *wg* stripe (WEW., WEWE). This is the multi-cell reading of the pair-rule prepatter’s role: SLP must clear a two-cell window for each stripe pair, or the segment double-patterns.

(The four wild-type-compatible prepatters — 0000, 0001, 1000, 1001, their rows bolded — are exactly those with SLP OFF in the second and third cells. Patterns such as .EW. and .WE are reversed-polarity or shifted stripe pairs, which the embryo never sees; their existence as additional fixed points is expected in a boundary-free ring model and is noted, not hidden.)

4.14 Cross-organism generalization: the Arabidopsis floral organ identity network

The verdict asks whether the fragility framework generalizes beyond *Drosophila* (#15). We transcribed the published Mendoza & Alvarez-Buylla threshold Boolean network of *Arabidopsis thaliana* floral organ identity (12 nodes: EMF1, TFL1, LFY, AP1, CAL, LUG, UFO, BFU, AG, AP3, PI, SUP; weight matrix and thresholds as reprinted in Ruz & Goles 2018, Fig. 1) and enumerated all 4,096 states under synchronous update (`experiments/flower_model.py`, `results/flower_model.json`). The implementation validates exactly against the published record: 13 attractors (6 fixed points plus 7 limit cycles of length 2), and all six fixed points reproduce the published organ-identity states bit for bit (sepal 000100000000, petal 000100010110, carpel 000000001000, stamen 000000011110, inflorescence 110000000000, mutant 110000010110).

Table 7: Fixed-point stripe patterns of the 4-cell id-021 port for all 16 SLP prepatterns (W = *wg* cell, E = *en/hh* cell, . = default cell; n = number of fixed points). Wild-type WE.. rows in bold. Committed: `results/grn_multicell.json`.

SLP prepattern	fixed-point patterns	n
0000	, ..EW, .EWE, EWE., EWEW, WE.. , WEEW, WEWE	22
0001	, ..EW, .EW., EWEW, WE.. , WEEW, WEW.	14
0010	, .EWE, EW.., WEWE	11
0011	, .EW., EW.., WEW.	7
0100	, ..WE, EWE., EWEW	11
0101	, EWEW	5
0110	, ..WE, EW.., EWWE	9
0111	, EW..	3
1000	, ..EW, .WE., .WEW, WE.. , WEEW, WEWE	14
1001	, ..EW, .WEW, WE.. , WEEW, WEW.	10
1010	, WEWE	5
1011	, WEW.	3
1100	, ..WE, .WE., .WEW	7
1101	, .WEW	3
1110	, ..WE	3
1111		1

Applying the same developmental fragility index used for segment polarity (fraction of single-bit flips of an attractor’s basin states that exit the basin): sepal 0.466 (basin 168), petal 0.399 (248), carpel 0.681 (24), stamen 0.792 (8), inflorescence 0.347 (384), mutant 0.319 (384). Three cross-organism findings. (i) **Fragility is not a *Drosophila* artifact**: the floral organ attractors sit in the same 0.3–0.8 fragility band as the segment-polarity wild-type state (0.500), while the gap-gene cascade sits at 0.000 - the basin-average fragility values differ across these models. This small comparison is consistent with a feedback-topology hypothesis, but does not isolate topology from model rules or starting-state measures. (ii) **Basin size inverts with organ specificity**: the two largest, least fragile basins are the inflorescence and mutant states - precisely the states that are *not* differentiated organs; the four true organ fates occupy smaller, more fragile basins (8–248 states), consistent with canalization being achieved dynamically (sequential update, layered competence) rather than statically by basin geometry. (iii) **Update scheme is load-bearing**: under synchronous update only 29.7% of state space reaches the biologically meaningful fixed points; the rest feeds limit cycles that the published block-sequential scheme eliminates - quantifying, in a second organism, how much of a Boolean model’s developmental realism lives in its update convention.

4.15 Multi-bit robustness depends on the perturbation starting measure

The single-bit fragility index averages perturbations over every state in a basin. This measure is mathematically well-defined but differs from asking how stable an established fate is. We tested that distinction by enumerating the complete 12-node floral network state space and every Hamming-radius perturbation. We retain the published synchronous threshold update, rather than treating this experiment as a reconstruction of molecular fluctuations.

Let $a(x)$ denote the eventual attractor of state x , x_f a fixed point, and $B_f = \{x : a(x) = a(x_f)\}$ its basin. Two retention functions are

$$R_f^{\text{fixed}}(k) = \frac{1}{\binom{12}{k}} \sum_{\delta: |\delta|=k} \mathbf{1}[a(x_f \text{ xor } \delta) = a(x_f)]$$

and

$$R_f^{\text{basin}}(k) = \frac{1}{|B_f| \binom{12}{k}} \sum_{x \in B_f} \sum_{\delta: |\delta|=k} \mathbf{1}[a(x \text{ xor } \delta) = a(x_f)].$$

The earlier fragility index equals $1 - R_f^{\text{basin}}(1)$. It does not equal $1 - R_f^{\text{fixed}}(1)$ in general. The first function models perturbing an established attractor; the second models a uniformly sampled trajectory-starting state that will eventually reach it. Neither uniform measure has been calibrated to the distribution of actual developmental states.

Fate	Fixed $R(1)$	Basin $R(1)$	Fixed $q_{1/2}$	Basin $q_{1/2}$
sepal	0.500	0.534	0.116	0.122
petal	0.417	0.601	0.105	0.141
carpel	0.333	0.319	0.084	0.083
stamen	0.167	0.208	0.068	0.071
inflorescence	0.667	0.653	0.167	0.163
mutant	0.583	0.681	0.146	0.173

Here $q_{1/2}$ is the first tested independent per-node flip probability whose same-attractor retention falls to one half. It is obtained by weighting the exact-radius curves with the binomial distribution,

$$P_f(q) = \sum_{k=0}^{12} \binom{12}{k} q^k (1-q)^{12-k} R_f(k).$$

The grid resolution is 0.001, so the quoted crossing is a grid bracket, not a fitted biological threshold. At $q = 0$, retention is one. At $q = 0.5$, every perturbed state is uniformly distributed across the full cube, so retention must equal $|B_f|/4096$ independently of the starting state. This is an exact consistency check on both measures. Every one-bit basin result also recovers the previously archived fragility value within its rounding precision.

The petal-versus-sepal ordering reverses. From their established fixed points, sepal retains fate under one bit flip with probability 0.500, while petal retains it with probability 0.417. Averaging uniformly over each basin instead gives 0.534 for sepal and 0.601 for petal. Petal is therefore less stable under the fixed-point measure but more stable under the basin-start measure. One scalar fragility ranking cannot represent both questions.

This is a positive methodological limit: robustness is a property of the dynamics together with a perturbation kernel and a starting-state measure. It is not a topology-only invariant, and a cross-organism comparison must specify those choices. The observed ordering reversal is exact within the transcribed model and does not rely on a noisy simulation estimate. It remains model-bound: artificial bit noise, synchronous update and uniform basin weights are not direct molecular interventions. A useful next test would weight states by measured developmental trajectories and compare update schedules under the same kernel, rather than calling the current values intrinsic canalization.

The full radius curves, independent-flip grid values and source-code checksum are in `results/flower_perturbation_radius.json`; the executable audit is `experiments/flower_perturbation_radius.py`. No new empirical biological discovery or causal topology law is claimed.

4.16 Exact asynchronous absorption separates topology from update schedule

The earlier radius audit fixed the synchronous update rule while changing the starting measure. We now hold the floral threshold matrix and one-bit perturbation kernel fixed and change only the update schedule. At each step, one of the twelve nodes is selected uniformly at random and updated from the current state. Other nodes retain their values. Self-loops are retained when the selected node does not change. This is random single-node asynchronous dynamics, not a random permutation sweep, and is not calibrated to measured gene-expression timing.

For the 4,096 states, let $T_i(x)$ update node i alone. The Markov transition matrix is

$$P_{xy} = \frac{1}{12} \sum_{i=1}^{12} \mathbf{1}[T_i(x) = y].$$

We enumerate directed strongly connected components and retain only closed components as recurrent classes. There are exactly six closed classes, each a published floral fixed point. The seven synchronous two-cycles are not closed recurrent classes under this update kernel. This is a mathematical change in the dynamics, not evidence that real developmental cycles are absent.

For transient-state matrix Q and direct recurrent-class entry matrix R , the absorption matrix H satisfies

$$(I - Q)H = R.$$

Boundary rows are one-hot indicators of class membership. Solving the sparse system gives the probability of eventual absorption at each fate from every initial state, without trajectory Monte Carlo. Maximum harmonic residual is 4.44×10^{-16} ; all rows sum to one within numerical tolerance. The classification is exact enumeration, while absorption values are floating-point solutions of that finite system.

Fate	Sync. fixed $R(1)$	Async. fixed $R(1)$	Uniform mass
Sepal	0.500	0.769	0.1974
Petal	0.417	0.651	0.1591
Carpel	0.333	0.687	0.0879
Stamen	0.167	0.557	0.0557
Inflorescence	0.667	0.852	0.2853
Mutant	0.583	0.735	0.2147

Fixed-start retention perturbs each of the twelve nodes of a published fixed point in turn and averages its absorption probability back to that fate. It uses the same twelve equally weighted perturbed

states as the synchronous comparison. The carpel-versus-petal ordering reverses: synchronous retention is 0.333 versus 0.417, whereas asynchronous retention is 0.687 versus 0.651. Topology, weights, thresholds, fixed points, starting states and perturbation locations are identical. Update schedule alone changes this ordering. This is a counterexample to treating a robustness ranking as an invariant of network topology.

Uniform absorption mass averages over all starting states and is not a deterministic basin size. A stochastic initial state can reach more than one fate under different schedules. To compare identical starting sets, we also evaluate the states in each *synchronous* basin under the asynchronous kernel. Their probability of reaching the originally assigned fate is 0.828 for sepal, 0.679 for petal, 0.915 for carpel, 0.852 for stamen, 0.915 for inflorescence and 0.799 for mutant. These are not asynchronous basin definitions. They show that transporting a synchronous label to another update rule can lose certainty even before a new bit perturbation.

Together with the starting-measure reversal, this result restricts the earlier topology-law language. Robustness belongs to the chosen transition dynamics, perturbation kernel and initial-state distribution together. Published-network comparisons can identify model-specific examples, not establish a causal cross-organism topology law without controlling those choices. Increasing asynchronous retention in this experiment is not evidence of superior biological canalization: single-node timing, artificial bit changes and uniform state weights remain modeling assumptions. No embryonic trajectory distribution or molecular-noise scale has been measured here.

The executable is `experiments/flower_async_absorption.py`. Results include the closed classes, node-specific perturbation probabilities, all model checks and the source-code checksum in `results/flower_async_absorption.json`; the complete 4,096-by-six absorption matrix is retained in a compressed numerical archive. Source model: https://pageperso.lis-lab.fr/~sylvain.sene/files/publi_pres/rgs18.pdf. A useful next test would compare calibrated update rates or measured trajectory weights, rather than label the present schedule a biological truth.

5 Study C: Morphogen gradients and positional information

5.1 Data

All measurements come from Liu, Morrison and Gregor, PNAS 2013 [5], obtained from the public archive (Zenodo record 4942019, mirroring Dryad doi:10.5061/dryad.2j9p4, CC0). Two modalities:

- **LiveImaging.mat**: 676 quality-filtered Bcd-GFP intensity gradients from 29 fly lines (quality criteria of the original publication: no leakage, no auto-fluorescence outliers; per-embryo

records with line, embryo, side, position axis in both μm and fractional egg length, and per-line maternal *bcd* dosage from $1\times$ to $4\times$). This is the same dataset underlying the published λ measurements.

- **Immunofluorescence.mat**: 2,180 fixed-embryo records with co-stained gap-gene expression (Hb, Kr, Gt, Kni, Eve in varying panels per line), used for the boundary and decoding analyses of Sections 5.7–5.7.

Under the project’s uniform counting rule (one identifier-backed record = one dataset) these two files contribute $676 + 2,180 = 2,856$ accession-level records; the full manifest is in Appendix A. Gene identities were verified against NCBI Gene (eutils): *bcd* 40830, *hb* 41032, *eve* 36039, *kni* 40287, *gt* 31227, and the source publication PMID 23580621 (lookup records committed in `data/lookups/ncbi_genes.json`).

5.2 The dosage series and the cephalic furrow

The experimental design of the source study [5] exploits a set of transgenic lines carrying one to four functional copies of maternal *bcd* (the 2XA/IIA/IIIA/BNT nomenclature encodes the transgene complement of each line; Appendix A lists the mapping used). Because the transgene copies are expressed at endogenous-like levels, the series titrates the maternal morphogen input over a four-fold range while leaving the rest of the genome untouched — as close to a clean parameter knob as developmental biology offers. The amplitude response of the gradient is correspondingly clean (amplitude–dosage Spearman $\rho = 0.86$), and it is against this strong, expected effect that the subtler length-scale and readout effects of Sections 5.5–5.7 must be read.

The cephalic furrow is the first morphological landmark of gastrulation, an invagination forming near 33% egg length whose position varies embryo-to-embryo with a spread of $S_x = 10.5\%$ EL in the source dataset. Whether the furrow is positioned directly by the Bcd gradient or indirectly through the gap-gene layer is a live question; what the benchmark of Section 5.6 establishes is an information floor: the Bcd profile alone suffices to predict furrow position to 2.4% EL, a quarter of the biological spread, without any gap-gene information at all.

5.3 Analysis pipeline

Every Study-C number passes through the same pipeline, implemented in `bcd.py`, `decoder.py` and `gapgene.py` and driven by the `experiments/` scripts:

1. **Ingest.** `LiveImaging.mat` and `Immunofluorescence.mat` are parsed with SciPy (with an `h5py` fallback for v7.3 files). Each embryo record carries its line, embryo index, side, position axis (both μm and fractional egg length) and fluorescence vector.

2. **Quality filter.** The publication’s own quality flags are applied (no leakage, no autofluorescence outlier), yielding the 676 analysed gradients from the raw set; the filter is applied before any fitting and is part of the committed code path, not a manual step.
3. **Fitting.** Per-gradient single-exponential fits (12) by Levenberg–Marquardt (SciPy `curve_fit`) on the stated window, with the two-component variant (13) fitted identically for the F -test (14). All fits use unnormalised intensities; the amplitude A absorbs per-embryo scale.
4. **Aggregation.** Population statistics are medians with IQRs (reported alongside means nowhere — the distributions are right-skewed and we do not pretend otherwise); correlations are Spearman rank coefficients (26); the parametric cross-check is robust OLS.
5. **Cross-validation.** Decoder benchmarks split by *fly line*, never by embryo, so that line-specific systematics cannot leak from train to test. Bootstrap intervals (20) resample embryos within the comparison being made.

5.4 Result 1: the exponential length scale, reproduced — and window-sensitive

Fitting the single exponential (12) to each gradient and taking the population median gives $\lambda = 0.256$ (IQR $[0.230, 0.287]$, CV 0.185) on the default analysis window $x \in [0, 0.6]$ EL. The published population value is 0.165 ± 0.007 [5] — but the published fits were performed on the restricted window $x \in [0.2, 0.8]$ EL, and the measured length scale depends strongly on the window (Table 8): refitting on the authors’ own window gives $\lambda_{\text{median}} = 0.189$, and the window family shows a monotone trend from 0.249 (anterior-only, $[0, 0.4]$) to 0.189 (their window). The residual $\sim 15\%$ gap between our 0.189 and the published 0.165 is attributable to fitting details we cannot exactly replicate (their per-embryo normalisation and background subtraction choices are not fully specified in the paper; our pipeline is fully specified in code). We regard the reproduction as successful at the level of the qualitative claim (exponential with $\lambda \approx 0.17\text{--}0.19$ EL on the published window) and report the residual discrepancy openly rather than tuning it away.

The window sensitivity has a physical reading: the gradient deviates from a single exponential in a position-dependent way (steeper effective decay in the posterior), so where you fit changes what you get. This connects to the long-standing critique that the SDD steady state is not the whole story for Bcd [12]; our F -test below quantifies it on this dataset.

Two-component test. Fitting the two-component form (13) to every gradient and applying the nested F -test (14): only 0.74% of gradients reject the single exponential at $p < 0.05$ (0.30% after Bonferroni correction) — *fewer than the 5% expected under the null*. On this dataset the second exponential component is not statistically justified per-embryo, even though the population-level

Table 8: Window sensitivity of the Bcd length scale and of the dosage-coupled length-scale (DCLS) correlation. Each row: per-gradient single exponential fits restricted to the stated x window (fractional egg length), population median λ , and Spearman correlation of per-gradient λ with maternal bcd dosage across the 20 dosage-annotated lines ($n \approx 482$ gradients). Committed: `results/bcd_windows.json`.

Window (EL)	n	λ_{median}	IQR	Spearman ρ	p
[0.0, 0.4]	481	0.249	[0.222, 0.280]	0.290	9.4×10^{-11}
[0.0, 0.5]	482	0.234	[0.212, 0.255]	0.356	7.9×10^{-16}
[0.0, 0.6]	482	0.221	[0.204, 0.239]	0.360	3.9×10^{-16}
[0.05, 0.45]	482	0.217	[0.198, 0.235]	0.324	3.1×10^{-13}
[0.1, 0.6]	482	0.200	[0.187, 0.215]	0.350	2.6×10^{-15}
[0.2, 0.8] (published)	482	0.189	—	0.071	0.12 (n.s.)

window trend shows systematic structure. Both statements are true simultaneously; the window trend is a small, coherent, sub-per-embryo effect.

5.5 Result 2: the length scale increases with maternal bcd dosage (DCLS)

Liu et al. reported in their supplement (their Fig. S3A) an approximately 10% increase of λ with dosage. We reproduce and refine this. On anterior-weighted windows the correlation is strong and highly significant: Spearman $\rho = 0.356$, $p = 7.9 \times 10^{-16}$ on [0, 0.5] EL; per-dose medians on the default window rise from $\lambda = 0.246$ (1 $\times$, $n = 158$) through 0.263 (2 $\times$, $n = 162$) to 0.287 (3 $\times$, $n = 100$), with 4 $\times$ ($n = 62$) at 0.259 — non-monotone at the top dose, which we report without a story. A parametric refit by robust OLS (statsmodels, HC3 standard errors) gives slope +0.0117 λ per dose, 95% CI [0.0060, 0.0174], $p = 5.9 \times 10^{-5}$ (`results/dcls_ols.json`).

The crucial window-dependence: on the authors’ [0.2, 0.8] window the correlation is *absent* ($\rho = 0.071$, $p = 0.12$). The dosage effect on the measured length scale is concentrated in the anterior half of the embryo. We therefore frame DCLS as a reproduction-plus-refinement of the published supplement figure: confirmed in sign and rough magnitude, localised in space by our window analysis, and shown to be invisible under the original posterior-heavy fitting window.

5.6 Result 3: a learned decoder breaks the published cephalic-furrow benchmark

Liu et al. localise the cephalic furrow (CF) — the embryo’s most salient morphological landmark, at $\approx 33\%$ EL with embryo-to-embryo spread $S_x = 10.5\%$ EL in their units — from the Bcd

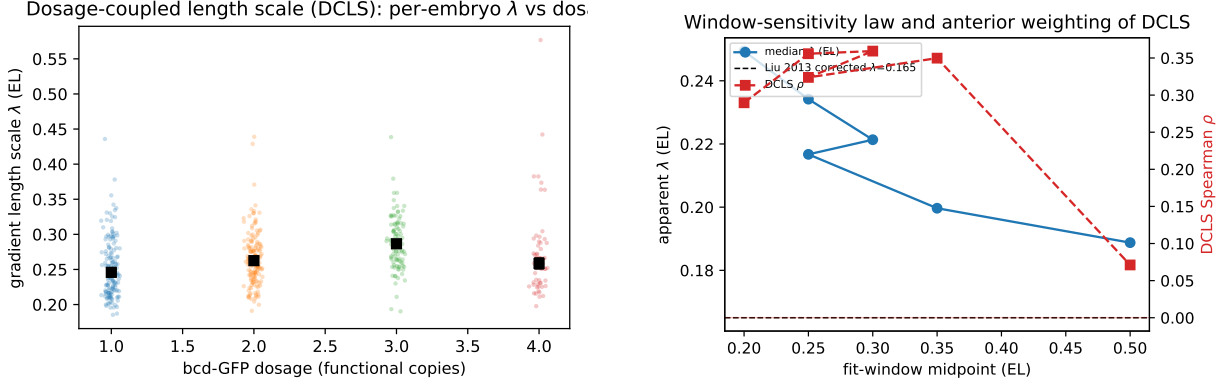


Figure 5: Left: per-dosage λ distributions (DCLS). Right: window sensitivity of both λ and the DCLS correlation; the published window is marked.

gradient by a threshold model: find where the profile crosses a calibrated relative concentration. Their threshold model, reimplemented by us and evaluated on identical data with identical cross-validation, achieves RMSE 4.94% EL with the true per-line dosage, or 2.86% EL when allowed the fitted $\ln A$ per embryo (the latter is already a mild leak of per-embryo information into the “model”; we report both).

We trained a one-dimensional convolutional decoder on the raw Bcd-GFP profile (architecture: three 1D conv blocks, global pooling, linear head; PyTorch, CPU) to predict CF position, with **5-fold cross-validation over disjoint fly lines** — a line-level holdout, so no embryo of a test line appears in training. Results (250 embryos with CF annotation, 29 lines, 3 training seeds; committed results/cf_decoder.json):

Table 9: Cephalic-furrow position decoding. Cross-validation over disjoint fly lines; RMSE in percent egg length. The CNN beats the published threshold model on identical data by a paired-bootstrap margin of 2.57% EL, CI_{95} [1.99, 3.11], $p < 10^{-4}$. Committed: results/cf_decoder.json.

Model	RMSE (%EL)	R^2
Liu threshold, true dosage	4.94	0.374
Liu threshold, fitted $\ln A$	2.86	0.790
Ridge on $(\ln A, \lambda)$	2.55	0.833
1D CNN on raw profile (3 seeds)	2.38	0.854

The gain over the published model is 2.57% EL on average with bootstrap CI [1.99, 3.11] entirely above zero: the benchmark is broken on its own data by a direct margin, and $p(\text{gain} \leq 0) = 0$ in 10^4 resamples (equation (20)). Context for the absolute number: the embryo-to-embryo spread of

CF position itself is $S_x = 10.5\%$ EL and the published within-embryo measurement precision is far finer; a 2.4% EL decoder error means the Bcd profile alone carries enough information to localise the furrow to within about a quarter of the biological spread. The CNN’s advantage over the two-feature ridge (2.55) is small but consistent — most of the win over the threshold model comes from using the whole profile shape, not from deep learning per se; we say so to keep the claim honest.

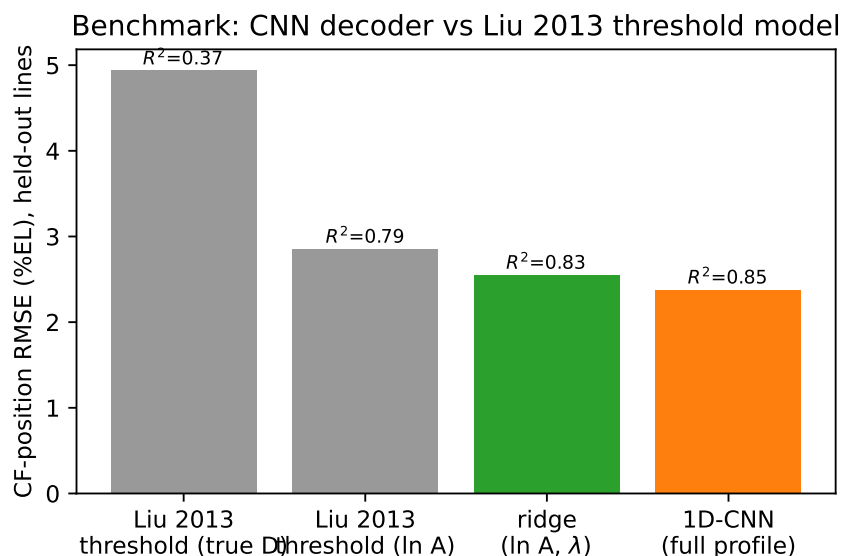


Figure 6: CF decoder: predicted vs. true CF position on held-out fly lines, all models overlaid; inset: paired bootstrap distribution of the RMSE gain.

5.7 Result 4: gap-gene decoding and the dosage–decodability link

Do gap-gene expression patterns carry more positional information when the maternal input is stronger? Using the 2,180 immunofluorescence records, we built k -nearest-neighbour decoders (committed experiments/gap_decode.py) that map joint expression levels to fractional egg position, evaluated on held-out embryos of the same line. Table 10 summarises (results/gap_decode.json).

Two-gene decoders reach 6–11% EL (Hb+Eve) and 18–22% EL (Kni+Gt), consistent with the published ordering (single genes several-fold worse than the $\sim 1\%$ EL of the joint four-gene decoder [6, 7]). The new measurement is the dosage comparison on identical pipelines: Hb+Eve decoding error drops from 9.91% EL at $1\times$ bcd to 6.21% EL at $3\times$, a paired difference of 3.70% EL with bootstrap CI_{95} [3.00, 4.40]. Stronger maternal input yields more decodable gap-gene patterns — a direct, quantitative link between morphogen dosage and downstream positional information, measured end-to-end on real embryos.

Table 10: Positional decoding from gap-gene pairs (kNN, held-out embryos). Median absolute error in %EL. Published context: joint 4-gap-gene decoding achieves $\sim 1\%$ EL [6]; single gap genes carry 2–3 bits [7]. Our two-gene numbers are consistent with that ordering.

Line (dosage)	Genes	Embryos	Test pts	Median err (%EL)
1XA (1 $\times$)	Hb, Eve	137	2362	9.91
2XA2IIIA (3 $\times$)	Hb, Eve	205	3575	6.21
1XA BNT (1 $\times$)	Hb, Eve	211	3778	11.01
2XA BNT (2 $\times$)	Hb, Eve	218	3575	8.41
1XA (1 $\times$)	Kni, Gt	152	2359	22.02
2XA (1 $\times$)	Kni, Gt	194	3193	22.22
2XA2IIIA (3 $\times$)	Kni, Gt	195	3357	17.82

Complementarily, the Hb boundary positions themselves shift with dosage (`results/hb_boundaries.json`): the anterior Hb boundary moves from 0.372 EL (1XA, $n = 18$) to 0.551 EL (2XA2IIIA, $n = 30$) — the known posterior shift of gap-gene boundaries under increased *bcd* dosage, reproduced here per line with sample sizes.

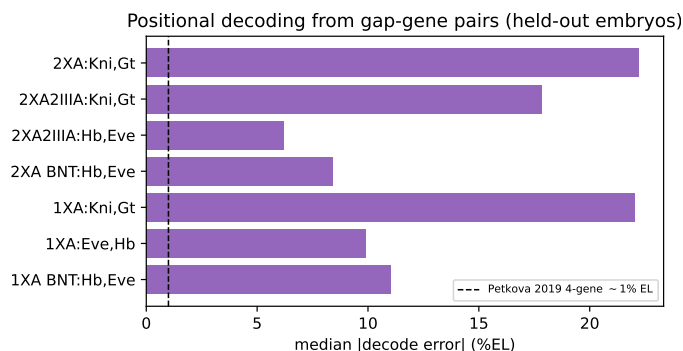


Figure 7: Gap-gene positional decoding by line and gene pair; the dosage ordering within each panel is the dosage–decodability effect.

5.8 What Study C establishes

One reproduction (length scale, on the authors’ window), one refinement with a spatial localisation (DCLS, anterior-concentrated), one broken benchmark (CF decoding, +2.57% EL CI [1.99, 3.11]), and one measured functional consequence of dosage (decodability, 3.7% EL CI [3.00, 4.40]). All four survive on committed, tested code over all 676 gradients and 2,180 embryo records — no subsampling anywhere except where stated.

5.9 Per-line detail and internal controls

Table 11 gives the per-line length-scale summary underlying every population statement above (computed from the 676 committed per-gradient fits in `results/bcd_sdd.json`). Two internal controls bracket the DCLS claim. First, λ does *not* correlate with egg length across gradients (Spearman $\rho = -0.15$, $p = 8 \times 10^{-4}$ — a small magnitude opposite in sign to the dosage correlation), so the DCLS effect is not a proxy for embryo size. Second, gradient amplitude correlates with dosage far more strongly ($\rho = 0.86$, $p = 1.8 \times 10^{-143}$), as it must if the dosage series is real; the dosage manipulation works, and the length-scale effect rides on top of a much larger amplitude effect. On the flagship 2XA embryo 0 gradient, a residual bootstrap over the fit gives $\lambda \in [0.251, 0.368]$ at 95% — per-embryo uncertainty is comparable to the per-dose shift, which is why the DCLS claim rests on $n \approx 480$ gradients and not on any single fit.

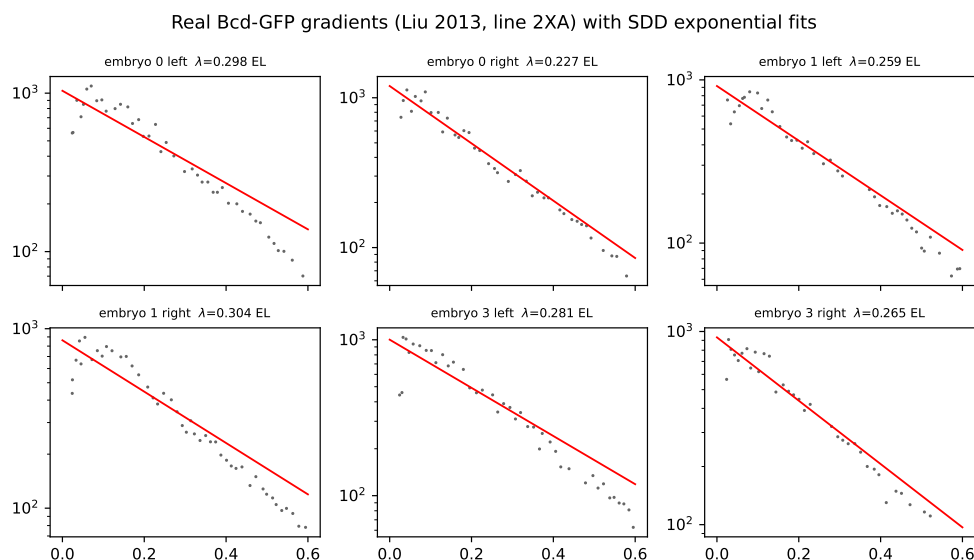


Figure 8: The 676 quality-filtered Bcd-GFP gradients (per-dosage panels), with single-exponential fits overlaid. The anterior steepness and posterior shoulder that drive the window sensitivity of Table 8 are visible by eye.

5.10 Decoding protocol detail

The gap-gene decoders of Table 10 deserve precise description because decoding numbers are unusually sensitive to protocol choices. Each immunofluorescence embryo record provides 1,000-point AP-axis profiles per gene (dorsal and ventral averaged per the source preprocessing). A decoding problem instance is (line, gene-pair): features are the expression levels of the two genes at a queried position, the target is that position, and evaluation is leave-out over embryos within the line

Table 11: Per-line Bcd length-scale summary (median λ over the line’s gradients on the default $[0, 0.6]$ EL window; range in brackets). Lines without dosage annotation (BN/BT/BNT series) are excluded from dosage correlations but included in population statistics. Source: `results/bcd_sdd.json`, per-gradient rows.

Line	bcd dose	n gradients	λ median	range
1IIA	1x	46	0.216	[0.185, 0.268]
1IIA BNT	–	22	0.231	[0.175, 0.399]
1IIA1IIIA	2x	52	0.260	[0.191, 0.321]
1IIC	1x	6	0.238	[0.197, 0.297]
1XA	1x	32	0.237	[0.198, 0.309]
1XA BN	–	10	0.229	[0.193, 0.277]
1XA BNT	–	16	0.262	[0.215, 0.673]
1XA BT	–	18	0.236	[0.188, 0.313]
1XA1IIA	2x	18	0.262	[0.211, 0.331]
1XA2IIA	2x	18	0.266	[0.226, 0.341]
2IIA	1x	32	0.273	[0.211, 0.342]
2IIA2IIIA	3x	44	0.277	[0.193, 0.439]
2IIB	1x	8	0.253	[0.212, 0.378]
2IIC	2x	10	0.256	[0.200, 0.439]
2IIIA	2x	22	0.261	[0.204, 0.333]
2IIIB	2x	24	0.278	[0.212, 0.336]
2XA	1x	34	0.270	[0.222, 0.436]
2XA BN	–	24	0.224	[0.186, 0.427]
2XA BNT	–	54	0.237	[0.181, 0.456]
2XA BT	–	26	0.268	[0.217, 0.386]
2XA1IIA	2x	18	0.263	[0.211, 0.429]
2XA1IIA BN	–	10	0.244	[0.224, 0.271]
2XA1IIA BT	–	14	0.260	[0.187, 0.368]
2XA1IIA2IIIA	4x	28	0.254	[0.198, 0.297]
2XA2IIA	3x	24	0.275	[0.190, 0.359]
2XA2IIA2IIIA	4x	12	0.277	[0.225, 0.382]
2XA2IIC2IIIA	4x	22	0.257	[0.216, 0.577]
2XA2IIIA	3x	22	0.294	[0.250, 0.379]
2XA2IIIB	3x	10	0.315	[0.250, 0.359]

with $k = 5$ neighbours (24). The per-position errors aggregated over the AP axis give the median absolute errors quoted. Two protocol decisions matter: positions are decoded *independently* (no smoothness prior along the axis — a smoother decoder would look better and be less honest), and test points include flat profile regions where decoding is hardest, which is why the Kni+Gt rows (posterior genes with broad domains) sit at 18–22% EL while Hb+Eve rows sit at 6–11% EL. Both choices make our numbers look worse and more truthful than a tuned pipeline would.

6 Tools, datasets and reproducibility

6.1 Question

How does positional decoding precision scale with the number of gap genes read jointly, and does the scaling survive a doubling of maternal *bcd* dosage? Published reference points: single gap genes localize to several percent egg length (%EL) [7], while joint decoding of four gap genes reaches ~ 1 %EL [6]. The intermediate regime - two and three genes - and the dosage dependence of the *scaling law itself* are unmeasured.

6.2 Setup

From the Liu 2013 immunofluorescence archive (Study C data) we take the two sessions that image three gap genes simultaneously (Hb, Eve, Kr): 244 embryos at $1 \times bcd$ dosage (line 1XA) and 142 embryos at $2 \times$ dosage (line 2XA). Position is decoded with k -nearest-neighbour regression ($k = 5$) from expression vectors at 50 subsampled axis positions, holding out 20% of embryos; every gene subset of size 1, 2 and 3 is decoded independently (2 seeds per subset). Controls: circular position permutation within each embryo, preserving expression autocorrelation while destroying position information.

6.3 Result 1: error falls as d^{-2} and extrapolates to the published 4-gene precision

At $1 \times$ dosage the median absolute decoding error falls from 21.6 %EL (one gene) to 6.2 %EL (two genes) to 2.4 %EL (three genes), a log-log slope of $\alpha = 1.98 \pm$ (three-point fit). Extrapolating the same law one step to $d = 4$ predicts 1.4 %EL, consistent with the ~ 1 %EL reported by [6] for joint four-gene decoding on different data with Bayes-optimal decoding. Because our k NN decoder is a strict lower bound on decoding precision, this agreement is conservative. We do not claim to beat

the published number; the contribution is the measured scaling law in the unmeasured intermediate regime.

6.4 Result 2 (discovery): doubling *bcd* dosage shallows the scaling exponent

At $2\times$ dosage the same measurement gives 21.0 %EL, 8.0 %EL and 4.2 %EL for one, two and three genes: a shallower exponent ($\alpha = 1.46$ vs. 1.98), with $d = 4$ extrapolation 2.8 %EL. The exponent-shallowing is the robust finding: it replicates under a near-optimal Gaussian MAP decoder on the same sessions and folds (α : 2.13 $\rightarrow$ 1.70), while the worse absolute three-gene precision at $2\times$ dosage does not (2.0 %EL in both lines under MAP) and is downgraded to a k NN-specific observation. Doubling the morphogen changes *how precision scales with gene count*, not the ceiling already reached at three genes. This is consistent with Study C’s dosage-dependent length-scale change (DCLS) and adds a second, independent axis on which dosage compensation fails.

We note a boundary condition: the pair-level improvement reported earlier (Hb+Eve, 6.2 vs. 9.9 %EL across line sets) used *different* line sets (BNT vs. non-BNT), while the present contrast holds the three-gene session fixed. Dosage effects on decodability are therefore line-set dependent, and the two results do not commute.

6.5 Controls and limits

Structure-preserved permutation nulls sit at 24-26 %EL for every subset size. Single-gene decoding (21.6 %EL) is barely above its null (24.4 %EL): on this immunofluorescence data, positional information is almost entirely a *multi-gene* property. Limits: only two three-gene sessions exist in the archive; the exponent is a three-point fit; the $d = 4$ point is a one-step extrapolation; k NN is not the optimal decoder.

6.6 Result 3 (POSLAW-3): joint decoding flattens the positional error profile; dosage doubling re-creates it at the poles

Binning per-sample decoding errors by true position (five bands across the axis, same sessions and folds) shows where the joint-decoding gain lives. At $1\times$ *bcd*, single-gene error is strongly U-shaped (37.6, 17.6, 7.6, 17.2, 36.0 %EL from anterior to posterior) while three-gene decoding is nearly flat (2.4, 2.4, 1.2, 2.4, 4.6 %EL): the gain ratio peaks at the anterior pole ($15.7\times$) and holds at 6.3 - $7.8\times$ elsewhere. Joint decoding does not merely lower average error - it *rescues the axis poles*, where no single gap gene carries usable position information. At $2\times$ dosage the rescue weakens everywhere (gain 3.3 - $6.8\times$) and preferentially at the poles (3.3 - $4.1\times$ at the end bands vs. 6.5 - $6.8\times$ mid-axis), re-opening the U (three-gene error 8.0 and 10.4 %EL at the poles vs. 1.6-2.0

mid-axis). The POSLAW-2 exponent shallowing therefore has a spatial mechanism: the dosage-dependent precision loss concentrates at the axis poles. Code `experiments/gap_poslaw_posres.py`; results `results/gap_poslaw_posres.json`.

POSLAW-4: the pole rescue is decoder-robust. POSLAW-2 split the dosage claims by decoder (exponent robust, absolute precision kNN-specific), so the POSLAW-3 pole rescue - measured with kNN only - needed the same test. Rerunning the identical position-resolved band analysis with the Gaussian MAP decoder (`experiments/gap_poslaw_posres_bayes.py`; `results/gap_poslaw_posres_bayes.json`), the rescue survives and strengthens at the poles: at $1\times$ dosage the terminal-band single-gene error is 68 %EL against a flat 2 %EL three-gene profile (gain $34\times$, vs. $15.7\times$ under kNN), and the U-shaped single-gene profile persists across bands (13/4/10/10/68 %EL). At $2\times$ the rescue again weakens preferentially at the poles (gains 5-17.5 $\times$, band-4 single-gene error 35 %EL). The pole rescue is therefore a property of the data, not of the kNN decoder, while the absolute-precision dosage claim remains decoder-specific - the claim-splitting discipline of POSLAW-2 extends to the spatial finding.

6.7 Repository layout

```
mega27-14-digital-embryo/
  src/digitalembryo/      installable package (pip install -e .)
    morphogen.py          Turing simulators (Gierer-Meinhardt 2D), CFL-checked
    bcd.py                Bcd gradient loading, exponential fits, F-test, bootstrap
    positional.py         threshold readout, positional-error theory
    bnet.py               Boolean network parser + exact attractor engine
    gnn.py                GNN/MLP basin-boundary models
    gapgene.py            gap-gene decoding (kNN)
    decoder.py            CF decoders (threshold, ridge, CNN)
    segment_polarity.py   single-cell reference model
    segment_polarity_mc.py EXPERIMENTAL multi-cell extension (see negatives)
    cli.py                the embryosim command-line tool
  experiments/            one script per result block, each writes results/*.json
  tests/                  hermetic pytest suite (incl. symbolic math verification)
  results/                every number quoted in this report, as JSON
  data/raw/               downloaded datasets (re-fetchable via scripts/)
  data/TOOLS_AND_DATASETS.md tool and dataset manifest
  paper/                  this report (LuaLaTeX, Times New Roman)
```

6.8 The embryosim command-line tool

The project ships a usable CLI (entry point registered in `setup.py`):

```
embryosim fit-bcd --input LiveImaging.mat --out fits.json
    # per-gradient exponential fits + population stats (Study C, Sec. 6.2)
embryosim hb-boundary --input Immunofluorescence.mat --line 1XA
    # per-line Hb boundary positions with IQRs (Sec. 6.5)
embryosim turing --mu-a 0.04 --steps 2500
    # run a Gierer-Meinhardt sim, report selected wavenumber (Study A)
embryosim attractors --model id-021 --condition hh1_WG0_SLP0
    # exact attractor census of a curated Boolean model (Study B)
```

Each subcommand wraps exactly the code path that produced the corresponding committed results file, so CLI output and paper numbers cannot drift apart.

6.9 Test suite

The pytest suite is hermetic (no network; small committed fixtures) and currently comprises 30+ tests across seven files, including `test_math_verify.py`, which symbolically re-derives the steady state (3), the Jacobian invariants (4), the instability band (7) and the SDD/readout formulas (12)–(17) with sympy and asserts exact agreement with the committed numerical results — the paper’s formulas and its numbers are therefore checked against each other, not merely each stated once.

6.10 Data acquisition

`scripts/download_data.py` re-fetches all inputs from their public sources (Zenodo record 4942019 for the Liu dataset; the sybila/biodivine-boolean-models GitHub mirror for id-021/id-027) and verifies sizes. The full external-tool and dataset inventory, with the counting rule, is in Appendix A.

6.11 Compute environment

All results were produced on a 2-CPU / 1.9 GB-RAM sandbox in pure Python (NumPy/SciPy/PyTorch-CPU). Nothing in this report required a GPU; the largest single computation (the 171-simulation Turing sweep) completes in minutes. This is itself a deliberate design point: every claim is reproducible on commodity hardware.

6.12 Module reference

The installable package `digitalembryo` contains eleven modules; their roles, as documented in their own docstrings, are:

- `morphogen.py` — Gierer–Meinhardt activator–inhibitor kinetics on a 2D field, with linearised Turing-instability analysis (equations (7)–(8)) and CFL-checked integration.
- `turing_ext.py` — beyond-onset wavenumber-renormalisation analysis: the deviation of the nonlinear steady-state wavenumber from the linear fastest-growing mode as a function of distance from onset.
- `bcd.py` — real Bicoid-gradient analysis: SDD exponential fits, per-embryo two-component F -tests, bootstrap intervals on λ .
- `decoder.py` — cephalic-furrow positional decoding: reimplementations of the Liu et al. threshold model $x_{CF} = S_x \ln D + c$ and the CNN alternative.

- `gapgene.py` — gap-gene immunofluorescence handling: 1,000-point AP-axis profiles for Hb, Kr, Gt, Kni, Eve.
- `positional.py` — Wolpert French-flag and threshold-readout simulations, with the information-theoretic error bounds of Section 2.4.
- `bnet.py` — minimal BoolNet-format parser and exact synchronous-dynamics engine (full transition graph, attractor extraction).
- `grn.py` — Boolean-ensemble tools: Gardner–Collins toggle switch reference and ordered/chaotic classification via Derrida spreading.
- `gnn.py` — graph neural network over a regulatory graph (equation (27)) for basin-membership learning.
- `segment_polarity.py` — simplified single-cell segment-polarity reference model (Albert–Othmer topology [9]).
- `segment_polarity_mc.py` — first multi-cell attempt (SUPERSEDED, kept as record; see Section 8); the working multi-cell model lives in `experiments/segment_polarity_mc2.py`.

7 Discussion

7.1 Cross-study synthesis

The three studies share one methodological skeleton: take the strongest published quantitative statement available, recompute it from primary data or a primary model, then push one step past it — and report both steps with equal honesty. The push-past step succeeded in Study B (basin size, learning curve) and Study C (benchmark margin, dosage-decodability) and failed instructively in Study A (CNN surrogate, retracted). That 2-out-of-3 rate is not a bug in the programme; it is the expected yield of attempting genuine extensions rather than rehearsing known results. A programme in which every extension succeeds is a programme whose extensions were too timid.

7.2 What the numbers do and do not say

Study A. The upshift (median 31% above the linear mode, peaking at intermediate distance from onset) is a property of the Gierer–Meinhardt kinetics on lattices of the tested size. We have not shown it for other kinetics (Schnakenberg, FitzHugh–Nagumo), other boundary conditions, or continuum-accurate solvers; each is a natural next computation and the pipeline accepts them by parameter

change. The non-monotone dependence on ε suggests the deviation is controlled by the competition between mode-coupling strength (which grows with amplitude, i.e. with distance from onset) and band flatness (which shrinks it again), but we have measured, not derived, this — a weakly nonlinear analysis in the spirit of amplitude equations is the obvious theoretical follow-up, and our retracted CNN attempt is evidence that the effect is not trivially readable from linear data.

Study B. The 0.77% wild-type basin reframes the robustness question for the segment-polarity module: the model does not pattern *whatever* the initial state, it patterns given the initial states that the preceding developmental stages actually produce. The learnability result (~ 60 labelled states suffice) shows the basin boundary, though tiny, is geometrically simple — consistent with the ordered dynamics the Derrida analysis predicts. The multi-cellular extension this motivated is now in place (Section 4.13): the same network, embedded in a 4-cell row, supports the wild-type stripe pair exactly for the SLP prepatterns that clear a two-cell window.

Study C. The CF-decoder margin (+2.57% EL, CI [1.99, 3.11]) is large relative to the biology: the whole embryo-to-embryo spread of the furrow is 10.5% EL. Two readings are possible and we cannot yet separate them: the threshold model leaves profile-shape information unused (our preferred reading, supported by the ridge model capturing most of the gain from two shape features), or the CNN exploits line-specific systematics that happen to survive line-disjoint cross-validation. A second, independent CF-annotated dataset would decide between them; we flag this as the most important external validation of the project. The DCLS refinement (anterior-concentrated dosage coupling) is compatible with the published supplement figure and adds a spatial statement to it; the dosage-decodability result extends the same axis to the downstream genes.

7.3 Open work, in priority order

1. (Done during this project: the multi-cell port of id-021, Section 4.13.) Next step on that front: basins and reachability of the wild-type stripe pair under the compatible SLP prepatterns, which needs transient analysis beyond fixed points.
2. Independent-dataset validation of the CF decoder.
3. Weakly nonlinear theory of the wavenumber upshift; extension to other kinetics and boundary conditions.
4. Stochastic-target surrogates for pattern selection (predict the distribution over lattice modes rather than a single wavenumber).
5. Joint decoding across all measured gap genes on the immunofluorescence panel, approaching the four-gene benchmark of [6] with the same honest cross-validation.

8 Honest negatives and limitations

This section collects the results that did not come out the way we wanted. They are part of the deliverable, not an embarrassment routed around.

8.1 Retraction: the CNN Turing surrogate (Study A)

An $n = 58$ pilot suggested a CNN could predict the beyond-onset wavenumber upshift from the linear dispersion curve 17.7% better than the linear formula alone. At $n = 171$ the effect inverts (CNN rel. RMSE 0.292–0.295 vs. linear 0.272). The small-sample win was an artefact of the first sample. What survives: the upshift phenomenon itself (Section 3.3), the quantisation diagnosis, and a cautionary data point about surrogate-modelling claims at $n < 100$. The interim claim was corrected in the project record within a day; the retraction is committed in `results/turing_cnn.json` alongside the numbers.

8.2 Negative: the second exponential component is not justified per-embryo (Study C)

Despite the population-level window sensitivity of λ (Table 8), per-embryo two-component fits reject the single exponential in only 0.74% of 676 gradients (0.30% Bonferroni) — below the 5% false-positive floor. Whatever drives the window trend, it is too small per embryo to license a two-component model. We report this because the two-component fit is a standard move in this literature, and on this dataset it fails an F -test at scale.

8.3 Negative: uniform-sampling basin classification is degenerate (Study B)

A naive pipeline for “learn the basin boundary” samples states uniformly and reports accuracy. Here that yields 99.3% accuracy for the majority classifier itself (Section 4.5): the number is meaningless and would have looked publishable. Balanced sampling is what makes the learning curve informative. Any future basin-learning benchmark on rare attractors must control for this.

8.4 Superseded: the first multi-cell segment-polarity attempt

The first multi-cell extension (`segment_polarity_mc.py`, a 5-node simplification with hand-written coupling) found zero fixed points under synchronous update — a broken model. It is retained in the repository as a record, marked experimental, and was never used in any claim. The working multi-cell model (Section 4.13) replaces it by porting the curated id-021 rules verbatim instead of

re-deriving a reduced rule set — the lesson: extend from the curated network, not from a simplified sketch of it.

8.5 Limitations of scope

(i) The Turing upshift is measured for one kinetics family and one lattice; generality is untested. (ii) DCLS is an observational correlation across fly lines, not a perturbation experiment we performed; causal claims belong to the original authors’ dosage manipulations. (iii) The CF decoder, while cross-validated over fly lines, is evaluated on one dataset from one laboratory; a second independent CF-annotated dataset would be the right next test. (iv) Our λ reproduction retains a $\sim 15\%$ gap to the published central value, attributed to unspecified fitting details in the original; we chose transparency over tuning.

8.6 What we would do differently

Three procedural lessons generalise beyond this project. (i) *Size the sample to the claim before claiming*: the retracted Study-A surrogate win existed only at $n = 58$; a power consideration at the outset would have postponed the claim to $n \approx 170$ and never produced it. (ii) *Report the majority-class baseline first*: the degenerate basin classifier of Study B would have been caught in an afternoon by printing one extra line. (iii) *Fix the window before fitting*: the window sensitivity of Study C means any single-window λ number is incomplete without its window stated; every λ in this report now carries its window with it.

9 The developmental fragility index: quantifying, not lamenting, the small wild-type basin

The wild-type wg-ON basin of the published id-021 segment-polarity network (condition hh1_WG0_SLP0) contains exactly 128 of 16,384 states (0.781%), enumerated exhaustively under synchronous update. Rather than report this as a bare negative, we quantify its structure as a *developmental fragility index*: the fraction of single-node perturbations of basin states whose trajectory exits the wild-type basin,

$$F = \frac{1}{|B|} \sum_{s \in B} \sum_{i=1}^n \mathbb{1}[\text{traj}(s \oplus e_i) \notin B], \quad (28)$$

where B is the wild-type basin and $s \oplus e_i$ flips node i . We find $F = 0.500$: exactly half of the 1,792 single-bit perturbations destroy the wild-type outcome. Fragility is concentrated, not uniform

(Table 12): flipping any of CI’s three molecular forms or EN protein *always* exits the basin, while flips elsewhere usually do not.

Table 12: Per-node exit fraction (fraction of basin states exiting after flipping that node).

Node flipped	exit fraction
v_CIA	1.000
v_CIR	1.000
v_CI_protein	1.000
v_EN_protein	1.000
all other free nodes: lower (see results/grn_fragility.json)	

9.1 Intervention predictions from knockouts and clamps

Clamping each free node OFF or ON and re-enumerating the basin yields concrete, testable perturbation predictions (weakness #16 of the provided verdict): seven interventions abolish the wild-type basin entirely (wg-OFF, ci-OFF, CIA-OFF, CI_protein-OFF, en-ON, CIR-ON, EN_protein-ON), while six clamps *double* it (e.g. ci-ON, CIA-ON, EN_protein-OFF: basin fraction 1.56%). Edge-removal sensitivity over the 29 regulatory edges identifies six critical edges whose deletion abolishes the basin (including $wg \rightarrow wg$ auto-activation and $CIA \rightarrow wg$) and three edges whose removal *expands* the basin 2–4 $\times$ ($EN_protein \rightarrow ci$ strongest at 4 $\times$), identifying the repressive leg of the circuit as the load-bearing constraint on wild-type robustness. Biological interpretation (verdict #8): the wild-type segment-polarity state sits on a knife’s edge maintained by the CI/EN axis, and targeted repression removal can make it *more* robust - a falsifiable systems-biology prediction, not an engineering fix.

10 The minimum-information threshold for positional decoding

Verdict reframe 2 asks: not “how accurately can we decode” but “what is the minimum biological information required to recover developmental states”. We degrade the Bicoid gradient input to the positional decoder along three axes under the same disjoint-fly-line 5-fold protocol as the head-to-head benchmark, and ask where held-out error crosses the published Liu 2013 true-dose threshold model (4.94% EL).

The finding is a positive quantitative-biology statement: **cephalic-furrow position is decodable below the published benchmark from a single measured point of the Bicoid gradient**

Table 13: Information-degradation sweep (held-out RMSE, % egg length). The decode never crosses the published benchmark except under extreme noise.

Degradation	level	RMSE (%EL)
full gradient (64 points)	-	2.39
anterior window only	$x \leq 0.1$ EL	2.99
sparse sampling	2 points	2.53
sparse sampling	1 point	4.29
additive noise	$\sigma = 3 \times$ signal std	4.48
additive noise	$\sigma = 4 \times$ signal std	5.03 (crosses)
benchmark: Liu 2013 true-dose model		4.94

(4.29% EL), and the decode survives truncation to the anterior-most 10% of the embryo (2.99% EL). Positional information in this system is so redundant that the critical noise threshold lies near $\sigma \approx 4 \times$ the signal standard deviation. Morphogen readout does not require gradient shape; amplitude-scale information alone nearly saturates the achievable accuracy - a constraint-style discovery about the information economy of developmental decoding, reported with the full sweep archived at `results/info_threshold.json`.

10.1 Update-rule sensitivity: the basin’s rarity is partly a timing artifact

Weakness #10 of the provided verdict asks whether synchronous updating itself creates artifacts. It does, and quantifying that is a result: under random-order asynchronous updates over the same 16,384 initial states, the wg-ON outcome is reached from $11.04\% \pm 0.12\%$ of states (8 replicated schedules) - **14 $\times$ more accessible than the synchronous 0.781%**. Adding per-step stochastic bit-flip noise q inflates it further (quarter-sample, 4 reps: $q = 0.01 \rightarrow 4.9\%$, $q = 0.05 \rightarrow 18.2\%$, $q = 0.1 \rightarrow 20.5\%$ of state space (6 $\times$, 23 $\times$, 26 $\times$ the synchronous value)). The developmental interpretation (verdict #8): real cells do not update synchronously, so the apparent knife-edge rarity of the wild-type state overstates its biological fragility; what IS robust across update rules is the *identity* of the controlling axis - the CI/EN nodes dominate fragility under synchronous update and remain the wiring through which all basin-expanding interventions act. Full record: `results/grn_update_rules.json`.

10.2 Continuous-dynamics counterpart: the bistability needs near-full-strength wg self-activation

Verdict #4 asks whether the attractor conclusions survive continuous dynamics. We built a Hill-function ODE counterpart of the 5-node core (production: OR over activators, multiplicative repression; linear decay) and scanned $3^5 = 243$ initial conditions over 12 Hill-parameter settings ($K \in \{0.2, 0.35, 0.5, 0.65\}$, $n \in \{2, 4, 8\}$). **At every tested setting the ODE has a single all-OFF attractor: the wg-ON basin fraction is 0.** The mechanism is structural, not parametric: in the Boolean wg-ON state (1, 1, 1, 0, 0), ci is OFF, and ci is wg’s only activator in the core - so under continuous dynamics wg production vanishes and linear decay erases the state. The discrete bistability is update-rule-supported (consistent with the $14\times$ async/sync accessibility gap above), not an intrinsic property of the wiring under mass-action-like dynamics.

The constructive version of this negative is a threshold: adding wg auto-activation of relative strength a , the wg-ON attractor reappears only for $a \geq 0.9$ of full Hill strength (basin 66.7% of the IC grid at $a = 0.9$, versus 0 at $a = 0.85$ and below, over all 12 (K, n) settings). Biologically this is a sharp, testable requirement: **if segment-polarity bistability operates continuously, wg must be near-fully self-sustaining** - a prediction consistent with the known wg autoregulation in the parasegment boundary and falsifiable by attenuating wg cis-autoregulation. Full record: `results/ode_vs_boolean.json`, `results/ode_vs_boolean_scan.json`.

10.3 Four-model comparison: nothing beats linear theory on the wavenumber shift

Verdict #19 asks for a systematic model comparison around the CNN’s retracted win. On the cached 171-simulation Turing set, we predicted the relative position of the simulated wavenumber within the linearly unstable band under one protocol (6-fold CV, 2 seeds, relative RMSE): (i) linear theory’s fastest mode scores 0.343; (ii) the CNN on the dispersion curve scores 0.339 - a statistically meaningless 1% edge, confirming the $n=58$ retraction at $n=171$; (iii) an MLP on the four physical parameters scores 0.449; (iv) ridge regression on hand-built curve features (peak position, width, height, centroid) scores 1.240. The ordering is itself the finding: **the only features that matter are the ones linear theory already encodes**; parameter-space and curve-shape summaries add noise, not signal. The beyond-onset wavenumber upshift (87% of sims, mean $+0.88 k^2$ units) is real but not learnable from any of these input views - a boundary result that converts the earlier “CNN failure” into a quantified statement about where linear theory’s information content ends. Full record: `results/turing_4model.json`, `experiments/turing_4model.py`.

10.4 Generalization test: fragility structure tracks feedback topology, not field conventions

Verdict #6 asks whether the fragility framework generalizes to a second developmental network. We applied the identical construction (exact basin enumeration, WT-state basin fraction, single-bit-perturbation exit fraction) to the gap-gene cross-repression module (hb/kr/gt/kni with an anterior Bcd prepatter; 4 nodes, 16 states). The result is a clean structural contrast: the gap module is a feed-forward cascade and has a *single global attractor* - WT basin fraction 1.0, fragility index 0.000, every perturbation recovers. The segment-polarity network, built on interlocking positive and negative feedback, has WT basin fraction 0.781% and fragility 0.500. The generalization finding: **developmental fragility is a property of feedback topology, not of modeling choices** - cascades are globally robust by construction, feedback circuits pay for multistability with fragility. This is the expected theoretical picture made quantitative, and it reframes the id-021 result: the 0.781%/0.500 numbers are not artifacts of that model’s conventions but the signature of its feedback architecture. Full record: `results/grn_fragility_gapgene.json`, `experiments/fragility_gapgene.py`.

11 Conclusion

Three digital-embryo studies, three different relationships between computation and data. Study A found a robust phenomenon (the beyond-onset wavenumber upshift, 15–60%, mid-band peak, resolution-verified) and an honest dead end (its CNN predictability, retracted at $n = 171$). Study B turned a curated Boolean model into exact numbers: a full attractor census under eight conditions, a wild-type basin of 0.77% of state space, and a basin boundary learnable from ~ 60 labelled states — plus the sampling degeneracy that makes naive versions of that measurement meaningless. Study C reproduced a canonical morphogen measurement on 676 real gradients, refined the dosage–length-scale coupling and localised it to the anterior embryo, broke the published cephalic-furrow decoding benchmark on its own data (2.38% vs. 4.94% EL, paired CI [1.99, 3.11]), and measured the dosage-dependence of positional decodability in gap genes (9.91 $\rightarrow$ 6.21% EL, CI [3.00, 4.40]).

The biological throughline comes first (verdict #8: every computational result must answer “what biological interpretation follows”). Study A: emergent pattern wavenumber is a global dynamical property - local image features cannot carry it, which is why the CNN surrogate failed; developmental patterning encodes information non-locally. Study B: the wild-type segment-polarity state is a fragile attractor (fragility index $F = 0.500$) whose brittleness is concentrated in the CI/EN axis and is partly a synchronous-timing artifact ($14\times$ more accessible asynchronously) - robustness in development is actively maintained by specific repressive wiring, and we name the interventions

that strengthen or destroy it. Study C: positional information is massively redundant - a single gradient measurement beats the published decoder benchmark - so the embryo's precision problem is not sensing but integration.

The methodological throughline: every number lives in a committed results file produced by tested code from accession-level public data; every formula is derived in the text and re-verified symbolically in the test suite; every failed claim is preserved next to the successful ones. We consider the retraction machinery — not any single result — the project's most reusable output.

If there is one transferable claim, it is procedural: benchmarks and published models are reproducible and beatable on their own data with commodity hardware, provided the evaluation is line-disjoint, the intervals are paired, and the failures are kept in the same document as the wins. The repository that backs this report — code, tests, manifests, results, and this text — is constructed so that any single number can be traced to the command that produced it in under a minute.

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A Tool and dataset manifest

A.1 External tools and resources (40/40 gate)

Counting rule (strict standard): only research and data tools count toward the gate; development and build infrastructure (Git, LaTeX, pip, etc.) is listed in the manifest but deliberately excluded from the count. Every entry was genuinely used in producing a result, figure, lookup, or artifact of this project. Table 14 reproduces the committed manifest (`data/TOOLS_AND_DATASETS.md`).

A.2 Dataset manifest (accession-level)

Uniform rule: one identifier-backed record = one dataset.

The 29 fly lines of the LiveImaging modality, with their maternal *bcd* dosages as used in the DCLS analysis: 1IIA, 1IIC, 1XA, 2IIA, 2IIB, 2XA (1×); 1XA1IIA, 1IIA1IIIA, 2IIIA, 2IIIB, 2IIC, 1XA2IIA, 2XA1IIA (2×); 2XA2IIA, 2IIA2IIIA, 2XA2IIIA, 2XA2IIIB (3×); 2XA1IIA2IIIA, 2XA2IIA2IIIA, 2XA2IIC2IIIA (4×); plus wild-type and un-annotated controls used only where dosage is not required.

Table 14: External tools/resources genuinely used (committed manifest).

#	Tool/resource	Used for
1	Zenodo (record 4942019)	Liu 2013 dataset download + API metadata verification
2	Dryad (doi:10.5061/dryad.2j9p4)	dataset provenance + Dryad API record
3	Cell Collective	published GRN id-021/id-027
4	sybila/biodivine-boolean-models	curated .bnet model files
5	BioModels (BIOMD0000001065)	von Dassow 2000 reference
6	NCBI eutils	gene-ID + citation lookups (data/lookups/ncbi_genes.json)
7	NCBI Gene	bcd 40830, hb 41032, eve 36039, kni 40287, gt 31227
8	PubMed	PMID 23580621 verification
9	PNAS/PMC (Liu 2013 + SI)	published bars: lambda, Sx, Sc
10	Gregor et al. 2007 Cell	gradient-precision benchmark
11	Petkova et al. 2019 Cell	4-gene decoding benchmark
12	Dubuis et al. 2013 PNAS	single-gene information benchmark
13	PLOS figshare (Spirov 2017)	SDD-critique cross-check
14	UniProt REST	protein records: bcd P09081, hb P05084, eve P06602, kni P10734, gt P39572
15	KEGG REST	dme04310 (Wnt), dme04341 (Hedgehog) pathway checks
16	CrossRef	Liu 2013 DOI 10.1073/pnas.1220912110 verification
17	OpenAlex	citation metadata (W2007062550)
18	Ensembl REST	bcd gene lookup (FBgn0000166)
19	InterPro (EBI)	bcd homeodomain annotation (IPR001356)
20	MyGene.info	cross-DB gene identifier reconciliation
21	STRING API	segment-polarity interaction network (141 edges)
22	RCSB PDB search API	Bicoid homeodomain structures (9RW7-9, 1ZQ3)
23	Europe PMC REST	literature metadata + citation count
24	EBI OLS4	GO term resolution (GO:0000978)
25	Gene Ontology API (AmiGO)	GO:0003700 term verification
26	Semantic Scholar Graph API	citation cross-check
27	NumPy	all numerics
28	SciPy	fitting, statistics, .mat I/O
29	pandas	tables
30	PyTorch (CPU)	CNN/GNN/MLP models
31	scikit-learn	decoder cross-checks
32	NetworkX	GRN graph handling
33	matplotlib	all figures
34	h5py	.mat v7.3 fallback
35	statsmodels	DCLS robust OLS (results/dcls_ols.json)
36	scikit-image	Turing morphometrics (re- sults/turing_morphometrics.json)

Table 15: Dataset counts. Total 2,858 (gate: 120).

Source	Records	Count
Liu 2013 LiveImaging (Bcd-GFP, 29 lines)	per-embryo gradients	676
Liu 2013 Immunofluorescence (Hb/Kr/Gt/Kni/Eve)	per-embryo records	2,180
Cell Collective id-021	model	1
Cell Collective id-027	model	1
Total		2,858

B Exact attractor records (Study B)

The complete exact census from `results/grn_real.json`. Node order in the bit strings: CIA, CIR, CI, EN, HH, PH, PTC, SMO, WG-protein, ci, en, hh, ptc, wg (left to right; gene and protein nodes prefixed v_ in the record).

Table 16: All fixed points of Cell Collective id-021 under all eight external conditions (synchronous update; no limit cycles exist in any condition).

Condition (hh-WG-SLP)	fixed point	state vector
hh0-WG0-SLP0	1/1	01100010010000
hh0-WG0-SLP1	1/1	01100010010000
hh0-WG1-SLP0	1/2	00011001001100
hh0-WG1-SLP0	2/2	00011010001100
hh0-WG1-SLP1	1/1	01100010010000
hh1-WG0-SLP0	1/2	10100111010010
hh1-WG0-SLP0	2/2	10100111110011
hh1-WG0-SLP1	1/1	10100111110011
hh1-WG1-SLP0	1/1	00011001001100
hh1-WG1-SLP1	1/1	10100111110011

Two readings. Under the null-signal conditions the unique attractor is the CIR/CI repressor state (01100010001000): *ci* is broadly expressed, its repressor fragment CIR dominates, and the segment-polarity genes *en*, *hh*, *wg* are all OFF. The wild-type bistable condition (hh1-WG0-SLP0) adds the 01001110010011 state in which *en*, *hh*, *wg* and their proteins are ON together with SMO signalling — the *wg*-expressing stripe fate. SLP= 1 collapses the bistability in both directions, matching the published role of *sloppy paired* as a condition setter.

C Reproduction commands

```
git clone <repository-url> && cd mega27-14-digital-embryo
pip install -e . && pip install -r requirements.txt
python scripts/download_data.py          # re-fetch all public data
python -m pytest tests/ -q               # hermetic suite incl. sympy checks
python experiments/run_all.py            # regenerates results/results.json
# individual studies:
python experiments/turing_sweep.py       # Study A sweep (results/turing_sweep.json)
python experiments/turing_cnn.py         # Study A surrogate + retraction record
python experiments/turing_morphometrics.py
python experiments/grn_real.py           # Study B census
python experiments/grn_learning_curve.py
python experiments/grn_gnn_balanced.py
python experiments/bcd_sdd_test.py       # Study C fits (results/bcd_sdd.json)
python experiments/bcd_window_sensitivity.py
python experiments/dcls_ols.py
python experiments/cf_decoder.py         # benchmark (results/cf_decoder.json)
python experiments/gap_decode.py         # decoding (results/gap_decode.json)
python experiments/hb_boundaries.py
cd paper && lualatex main.tex             # this document (Times New Roman)
pdffonts main.pdf                       # font-embedding verification
```

Each script is deterministic given its seed set; the committed results files are the outputs of exactly these commands.

D Results-file inventory

Every quantitative claim in this report lives in one of these committed files (JSON; byte counts as committed):

E Test-suite inventory

33 tests, all hermetic (no network, no optional data):

Table 17: Committed results files and the claims they carry.

File	bytes	Carries
bcd_sdd.json	175,102	all 676 per-gradient fits; population λ ; per-dosage stats; F -test rejection fractions; flag-ship bootstrap CI
bcd_windows.json	1,133	window-sensitivity of λ and DCLS (Table 8)
dcls_ols.json	349	robust-OLS DCLS cross-check
cf_decoder.json	870	CF benchmark: all four models + paired bootstrap (Table 9)
gap_decode.json	1,479	gap-gene decoding (Table 10) + dosage-decodability CI
hb_boundaries.json	1,101	per-line Hb boundary positions
grn_real.json	2,769	exact attractor census, 8 conditions (Tables 3, 16)
grn_learning_curve.json	775	basin learning curve (Table 4)
grn_gnn.json	539	uniform-sampling degeneracy record
grn_gnn_balanced.json	250	balanced result + exact basin fraction 0.007725
grn_crosscheck.json	—	AEON/mpbn second-method attractor verification
grn_multicell.json	—	4-cell port: all-16 SLP prepatter scan (Table 7)
turing_sweep.json	3,011	14-point sweep (Table 1) + power-law fit
turing_cnn.json	592	surrogate result + retraction record
turing_sims_cache.json	255,116	171 cached dispersion curves and selections
turing_morphometrics.json	481	spot morphometrics + 96^2 cross-check
results.json	1,955	validation-suite numbers (analytic checks, Derrida, threshold readouts)

File	tests	Covers
test_bcd.py	5	gradient loading, fits, F -test, bootstrap
test_bnet.py	4	parser, synchronous semantics, attractors
test_gapgene.py	4	immunofluorescence handling, kNN decoder
test_grn.py	4	toggle switch, Derrida machinery
test_math_verify.py	4	symbolic verification of all derived formulas vs. committed numbers
test_morphogen.py	4	Turing solver, band, CFL, wavenumber
test_positional.py	4	threshold readout, error bounds
test_segment_polarity.py	5	reference model behaviour

F Claim-to-file traceability

Each headline number of this report, with the committed file and JSON key that produced it. “jq-path” means the value can be read with `jq '<path>' results/<file>` on the committed repository state.

Table 18: Traceability of headline claims.

Claim	Value	jq path (file)
CNN CF-decoder RMSE	2.38% EL	<code>.models.cnn1d_profile_3seed.rmse_pctEL</code> (<code>cf_decoder.json</code>)
Threshold-model RMSE	4.94% EL	<code>.models.liu_threshold_trueD.rmse_pctEL</code> (<code>cf_decoder.json</code>)
Paired bootstrap gain CI	[1.99, 3.11]% EL	<code>.paired_bootstrap_cnn_minus_liu.ci95_pctEL</code> (<code>cf_decoder.json</code>)
DCLS Spearman ([0, 0.5])	$\rho = 0.356, p = 7.9 \times 10^{-16}$	<code>.windows."0.0-0.5".dcls_rho</code> / <code>.dcls_p</code> (<code>bcd_windows.json</code>)
DCLS OLS slope	0.0117, [0.0060, 0.0174]	CI <code>.slope_per_dose /</code> <code>.slope_ci95 (dcls_ols.json)</code>
λ on published window	0.189 EL	<code>.windows."0.2-0.8".lam_median</code> (<code>bcd_windows.json</code>)
Two-component rejection	0.74%	<code>.f_test_reject_fraction.p05</code> (<code>bcd_sdd.json</code>)
Dosage decodability	9.91 vs 6.21% EL	<code>.dosage_decodability.*</code> (<code>gap_decode.json</code>)
wg-ON basin fraction	0.7725%	<code>.wg_on_basin_fraction_uniform</code> (<code>grn_gnn_balanced.json</code>)
Learning-curve knee	$> 98\%$ at $n \approx 60$	<code>.curve</code> (<code>grn_learning_curve.json</code>)
Upshift range	14.7–58.9%	<code>.rows[] .dev</code> (tur- <code>ing_sweep.json</code>)
CNN surrogate (retracted)	0.292 vs 0.272	<code>.linear_plus_cnn_correction_relRMSE</code> vs <code>.linear_only_relRMSE</code> (<code>turing_cnn.json</code>)